

FLIGHT DYNAMICS

Robert F. Stengel

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For Frank and Ruth

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Preface

Flight Dynamics takes a new approach to the science and mathematics of aircraft flight, unifying principles of aeronautics with contemporary systems analysis. It is a text and reference book for upper-level undergraduates and first-year graduate students as well as practicing engineers. While the book presents traditional material that is critical to understanding aircraft motions, it does so in the context of modern computational tools and multivariable methods. Particular attention is given to models and techniques that are appropriate for analysis, simulation, evaluation of flying qualities, and control system design. Bridges to classical analysis and results are established, and new territory is explored. The book is intended for readers with a broad background and interest in engineering and science. It begins with basic principles of trigonometry, calculus, linear algebra, and dynamics, and it continues with advanced topics, algorithms, and applications.

Dynamic analysis has changed dramatically during recent decades, with the introduction of powerful personal computers and scientific programming languages. Analysis programs have become so pervasive that it can be assumed that all students and practicing engineers working on aircraft flight dynamics have access to them. Therefore, this book presents the principles, derivations, and equations of flight dynamics with frequent reference to computed examples, and it includes MATLAB code for six-degree-of-freedom simulation and linear system analysis as on-line supplements.

Because it uses common notation and does not assume a strong background in aeronautics, *Flight Dynamics* is accessible to a wide variety of readers. Introductions to aerodynamics, propulsion, structures, and the atmospheric and gravitational environment precede the development of the aircraft's dynamic equations. *Système International* (SI) units are used for the majority of the text. These units take better advantage of decimal arithmetic than do U.S. Customary Units, the majority of world industry

and academe uses them, and virtually all high-school and college students learn physics, chemistry, and mathematics with SI units.

Bibliographic reference lists are presented at the end of each chapter and in Appendix E. Not all the books, reports, and papers are explicitly cited in the accompanying chapters. The lists are intended as resources for aerodynamic data and analytical results that are related to the associated text. Of course, many of the references did contribute directly to the book, and I am grateful to all of the authors for the insights that they provided.

Flight Dynamics is organized in seven chapters and an epilogue. “Introduction” (Chapter 1) presents a summary of the important elements of the aircraft and its systems; examples of about a dozen aircraft ranging from an uninhabited air vehicle, through general aviation and transport aircraft, to combat aircraft, trans/supersonic aircraft, and the space shuttle; and an overview of the mechanics of flight. “Exploring the Flight Envelope” (Chapter 2) describes the flight environment and the essential ingredients for steady flight. “Dynamics of Aircraft Motion” (Chapter 3) provides comprehensive coverage of the nonlinear equations of motion, expressed in various coordinate systems and with alternative angular measures (direction cosines and quaternions). “Methods of Analysis and Design” (Chapter 4) addresses the fundamental equations used for both classical and modern analysis of aircraft stability and control, including stochastic methods, and it presents introductions to aeroelasticity, flying qualities criteria, and flight control system design. “Longitudinal Motions” (Chapter 5) presents motions in the vertical plane in a novel fashion, building insight with reduced-order linear models derived from the complete set. “Lateral-Directional Motions” (Chapter 6) parallels the previous chapter, describing rolling and yawing motion with increasingly complex models and subjecting each to a series of analyses. “Coupled Longitudinal and Lateral-Directional Motions” (Chapter 7) goes into considerable detail about the ways in which motions interact, using bifurcation analysis, linear analysis, and nonlinear simulation, and it introduces several approaches to nonlinear control system design that are essential for high-angle-of-attack flight. The “Epilogue” briefly discusses the implications of the book.

One feature of the book is the inclusion of a nonlinear, six-degree-of-freedom simulation model for a generic business jet aircraft, which can be found on line at <http://www.pupress.princeton.edu/stengel>. The model is used for most examples, beginning in Chapter 2. The inertial and aerodynamic properties of the model are representative of business jets from subsonic to high-transonic airspeed and for angles of attack up to 90 deg. Together with the MATLAB nonlinear simulation program FLIGHT, the model is used to demonstrate motions across the spectrum of flight,

including effects of compressibility, stalling, spinning, and control nonlinearity (e.g., limit cycles and chaotic motion). FLIGHT can generate linear, time-invariant (LTI) dynamic models at arbitrary flight conditions. The LTI models can be subjected to various linear analyses in the MATLAB program SURVEY, including model order reduction through truncation or residualization, transient response, equilibrium response, controllability and observability, Bode, Nyquist, and Nichols plots, and root locus analysis. Longitudinal and lateral-directional motions can be analyzed separately or under fully coupled flight conditions. The FLIGHT and SURVEY programs are heavily commented, and they serve a tutorial role on their own.

While covering all the material in the book typically would require a two-semester course, a one-term undergraduate course can be based upon the book, leaving the more advanced topics for independent reading. Such a course would contain introductory material (Chapter 1), nonlinear equations of motion (Chapters 2 and 3), configuration aerodynamics (Chapters 2 and 3), aircraft performance (Chapter 2), linearized equations of motion and analysis (Chapter 4), longitudinal stability and control (Chapter 5), and lateral-directional stability and control (Chapter 6). Assignments may include the estimation of configuration aerodynamics for contemporary aircraft and various analyses of these aircraft conducted with FLIGHT, SURVEY, or student-generated computer code.

I am indebted to the many students, teachers, and colleagues with whom I have worked over the past years. It is (or should be) well known that teachers frequently learn at least as much from their students as the students learn from them. That is certainly the case for me, and I want to acknowledge the hundreds of graduate and undergraduate students who contributed to my education. Former teachers who introduced me to much of flight dynamics include Eugene Larrabee and Wallace VanderVelde at MIT, and Enoch Durbin, Dunstan Graham, David Hazen, Courtland Perkins, and Edward Seckel at Princeton. I also learned a great deal from many communications with Duane McRuer (Systems Technology), Albert Schy (NASA Langley), Malcolm Abzug (ACA Systems), and William Rodden (consultant).

Past and current associates provided a continuing flow of good ideas. These include Paul Berry and John Broussard at The Analytic Sciences Corporation, who had superb grasps of flight dynamics and control, as well as Princeton faculty and staff colleagues: Howard (Pat) Curtiss, Earl Dowell, David Ellis, Phillip Holmes, Jeremy Kasdin, Naomi Leonard, Michael Littman, Robert McKillip, Kenneth Mease, George Miller, and Barry Nixon.

Not to be forgotten are the research and consulting sponsors whose support was foundational. The NASA/FAA Joint University Program for

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Special thanks go to those who reviewed the book manuscript at various stages of its preparation. I am especially grateful to former student Jean-Paul Kremer, who read the manuscript critically and in remarkable detail, making countless discerning comments. Former students Dennis Linse, Axel Niehaus, and Muthathamby Sri Jayantha made a number of helpful suggestions, while Professor Haim Baruh (Rutgers) and Professor Eric Feron (MIT) perceptively reviewed the manuscript and provided valuable advice.

And finally, my warmest appreciation and love go to Pegi, Brooke, Christopher, and Brian, whose care, understanding, and support have been essential.

Chapter 1

Introduction

Flying has become so common that we tend to take many details of flight for granted. Nevertheless, flight is a complex process, involving the equilibrium, stability, and control of a machine that is both intricate and elegant in its design. All aircraft are governed by the same rules of physics, but the details of their motions can be quite different, depending not only on the shapes, weights, and propulsion of the craft but on their structures, control systems, speed, and atmospheric environment. This book presents the flight dynamics of aircraft, with particular attention given to mathematical models and techniques for analysis, simulation, evaluation of flying qualities, and control system design.

In this chapter, we introduce the basic components of configuration that are common to most aircraft (Section 1.1) and provide illustrative examples through descriptions of contemporary aircraft (Section 1.2). Notation that is used throughout the book is presented in Section 1.3, with an introductory example based on the flight of a paper airplane.

1.1 Elements of the Airplane

Aircraft configurations are designed to satisfy operational mission and functional requirements, with considerable consideration given to cost, manufacture, reliability, and safety. All aircraft have structures that generate aerodynamic lift, mechanisms for effecting control, and internal spaces for carrying payloads. Most aircraft also have propulsion systems and associated propellant tanks, undercarriage for takeoff and/or landing, and navigation systems. The layout and interactions of all these components have major influence on an aircraft's motions [e.g., B-1, C-1, G-2, H-1, H-2, K-2, N-1, R-1, S-1, S-2, S-3, T-1, and W-1].

AIRFRAME COMPONENTS

Consider the Cirrus SR20 aircraft shown in Fig. 1.1-1. A single-propeller-driven aircraft, the SR20 has all the elements of a conventional aircraft:

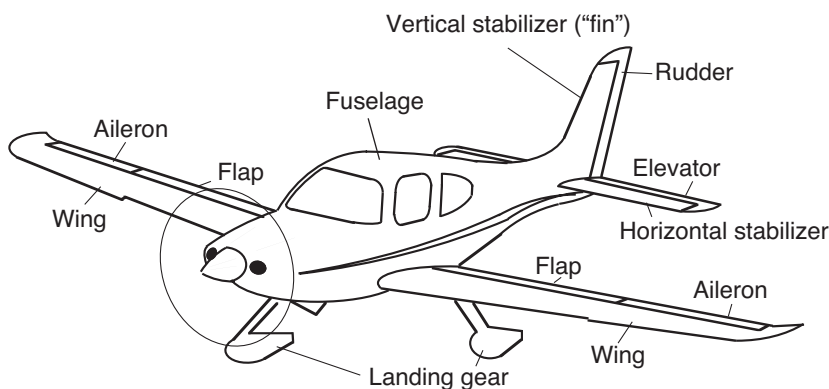


Fig. 1.1-1. Cirrus SR20 general aviation aircraft. (Courtesy of Cirrus Design Corp.)

wing, fuselage, horizontal tail, vertical tail, and control surfaces, as well as cockpit/cabin, engine, and landing gear. The *wing* provides the aircraft's largest aerodynamic force, used principally for supporting the vehicle's weight and changing the direction of flight. The wing's fore-aft location is near the vehicle's center of mass for good balance, and the wing itself is shaped to provide a large amount of lift with as little energy-absorbing drag as possible. The *fuselage* is the aircraft's principal structure for containing payload and systems. Its slender shape provides a usable volume for the payload and a mounting base for engine and tail components, with minimum drag and weight penalty.

Just like a weathervane, the *vertical tail* produces directional stability about an aircraft-relative vertical axis; it gives the aircraft a tendency to nose into the *relative wind* that results from forward motion. Similarly, the *horizontal tail* provides angular stability for rotations about an axis parallel to the wingspan. Aerodynamically, the vertical and horizontal tails are small wings whose lift acts through moment arms (between aerodynamic centers and the center of mass) to generate restoring torques that are proportional to angular perturbations. Collectively, the tail surfaces are called the *empennage*. Stability about the third (rolling) axis is a complex effect, dominated by the wing's vertical location on the fuselage and the upward inclination of the wing tips relative to the wing roots (the *dihedral angle*).

The SR20 has conventional control effectors: ailerons, elevator, rudder, flaps, and thrust setting. The *ailerons* are movable surfaces located near the wing tips that produce large rolling torques; the two surfaces are linked so that the trailing edge of one moves up when the trailing edge of

the other moves down. The *elevator* is a movable surface that extends across the trailing edge of the horizontal tail for angular control. In addition to regulating the aircraft's angle of attack,¹ which, in turn, governs the amount of lift generated by the wing, the elevator setting controls the pitch angle² when the wings are level, an important function during takeoff and landing. The *rudder* has a similar effect for yawing motions, controlling the sideslip angle³ for turn coordination and crosswind takeoffs and landings. It is a movable surface mounted on the trailing edge of the vertical tail.

The wing's trailing-edge *flaps* are movable surfaces mounted inboard of the ailerons for direct control of lift and drag during takeoff and landing. The left and right surfaces act in unison, deflecting downward from their cruising-flight (stowed) positions. Whereas the pilot exerts continuous control of the ailerons, elevator, and rudder, the flaps normally are adjusted to discrete settings that depend on the flight phase. Engine *thrust setting* is regulated only occasionally to achieve takeoff acceleration, climb, desired cruising conditions, descent, and landing sink rate.

A number of other subsystems have dynamic and control effects, including landing gear, trim tabs, leading-edge flaps, spoilers, speed brakes, thrust reversers, engine inlet shape and bleed air, jet exhaust deflection, and external stores. Functions may be combined or distributed, as for tailless aircraft or those possessing redundant control surfaces. These are configuration-dependent effects that are explored in later sections.

If each of the aircraft's elements performed only the functions described above and performed them perfectly, aircraft dynamics and control could be relatively simple topics; however, there are numerous complicating factors. Perhaps the greatest is that important physical phenomena are inherently *nonlinear*. The significance of nonlinearity is revealed in the remainder of the book, but the general notion is that doubling a cause does not always double the effect. For example, the wing's lift is linearly proportional to the angle of attack up to a point; then, the wing stalls and no greater lift can be achieved, even if the angle continues to increase.

A particular element (e.g., the wing) produces a primary effect (lift), but it may also produce secondary effects (drag and side forces, as well as pitching, yawing, and rolling moments). Because the components are in close proximity, the aerodynamic forces and moments that they

1. The *angle of attack* is the aircraft-relative vertical angle between the centerline and the relative wind. The *relative wind* is the velocity of the aircraft relative to the air mass through which it flies.

2. The *pitch angle* is the angle of the aircraft's centerline relative to the horizon.

3. The *sideslip angle* is the aircraft-relative horizontal angle between the centerline and the relative wind.

generate are interrelated. Hence, the airstream downwash and vorticity induced by the lifting wing have major effects on tail aerodynamics, thrust changes may upset pitch or yaw equilibrium, and so on. A good deal of useful analysis can be accomplished with single-input/single-output (scalar) models, but the actual dynamic system has multiple inputs and outputs, requiring a more comprehensive multi-input/multi-output (vector) approach.

There are challenging structural and inertial effects as well. The aircraft's structure must be lightweight for good overall performance, but it is very flexible as a result; consequently, the wing, fuselage, and tail deflect under air loads, and the changes in shape have both static and dynamic effects on equilibrium and motion. In modern aircraft, natural frequencies of significant vibrational modes may be low enough to be excited by and to interfere with aircraft maneuvers, leading to major problems to be solved in control system design. They also contribute to poor ride quality for the passengers and crew and to reduced fatigue life of the airframe. Although most aircraft possess mirror symmetry about a body-fixed vertical plane, inertial and aerodynamic coupling of motions can occur in asymmetric maneuvers. The loading of asymmetric wing-mounted stores on combat aircraft clearly presents a dynamic coupling effect in otherwise symmetric flight, and even a steady, banked turn couples longitudinal and lateral-directional motions.

PROPULSION SYSTEMS

All operational aircraft other than unpowered gliders and rocket-powered "aerospace planes" produce thrust by moving air backward with power from air-breathing engines. Power is generated either by the combustion of fuel and air in *reciprocating engines*, which use crankshafts to transform the linear motion of pistons moving in hollow cylinders to rotary motion, or in *turbine engines*, which produce rotary motion by the action of combustion gasses flowing through a finned wheel (i.e., a turbine). Power is most efficiently transformed to thrust by moving large masses of air at low speed rather than small masses of air at high speed; therefore, large-diameter *propellers* spinning at low rotational rate are indicated. Propellers are driven by either reciprocating engines or turbines; the latter arrangement is called a *turboprop* propulsion system if the propeller is of conventional design or a *propfan* if the propeller is designed for high-speed flow.

Propellers absorb unacceptable amounts of power as airspeed increases, inducing tip speeds that approach or exceed the speed of sound. The *turbofan* engine overcomes these problems by enclosing a smaller-diameter fan within a streamlined duct, allowing higher aircraft speeds but reducing

power conversion efficiency. The proportion of air that is merely accelerated by the fan compared to that which flows through the engine core is called the *bypass ratio*. The propfan mentioned earlier can be considered a turbofan with ultrahigh bypass ratio. With no bypass air accelerated by a fan, the turbine engine is called a *turbojet*—all the thrust-producing air mass passes through the combustion chambers and turbines. Turbojets were used in early supersonic⁴ aircraft, where the drag losses of the fan outweighed the fan's conversion efficiency. Advances in fan design have led to supersonic aircraft with turbofan engines. *Afterburning* produces increased thrust by burning additional fuel in the oxygen-rich exhaust of a turbojet engine, at the expense of reduced combustion efficiency.

For flight at a Mach number $\mathcal{M}$ of two or more, the air can be satisfactorily compressed for combustion by a properly designed inlet, so the compressor blades and the turbine that turns them are no longer necessary. The *ramjet* is an engine consisting of an inlet (which compresses the air and slows it to subsonic speed), combustion chamber, and exhaust nozzle. It proves to be the most fuel-efficient powerplant for $\mathcal{M} = 3\text{--}6$. Compressing the air increases its temperature; at even higher speeds, this temperature rise is excessive, challenging the structural strength of engine materials. The *supersonic-combustion ramjet* (or *scramjet*) provides one possible solution to this problem, slowing the air enough for adequate compression, though yielding a supersonic flow through the combustion chamber.

The *rocket* engine is capable of producing thrust at any Mach number and in a complete vacuum. Moreover, its structural weight per unit of thrust is less than that of any of the previously mentioned devices. The principal penalty is that the oxidizer as well as the fuel must be carried onboard the aircraft, increasing the size and weight of the vehicle. Consequently, the rocket is useful for launch to orbit, but it has very restricted roles to play in atmospheric flight, such as in short-term thrust augmentation for takeoff, maneuvering, or braking.

Future aircraft that operate over a wide range of Mach numbers will almost certainly use *variable-* or *multiple-cycle engines* (referring to the thermal cycles of combustion). For example, a *turboramjet* would operate as a turbojet at low speeds and a ramjet at high speeds, with the percentage of inlet air ducted directly to the afterburner increasing as Mach number increases. A rocket could accelerate an aircraft to $\mathcal{M} = 3$, possibly driving a turbine to compress inlet air for combustion, with a ramjet taking over

4. The *Mach number* $\mathcal{M}$ is the ratio of airspeed V to the speed of sound a in the surrounding air. The flow is *supersonic* when $\mathcal{M}$ is greater than 1 and *subsonic* when $\mathcal{M}$ is less than 1. Air moves faster or slower than the *freestream airspeed* over different parts of the aircraft. The *transonic region* begins at the freestream Mach number for which the flow reaches sonic speed over some part of the aircraft and ends when the entire flow is supersonic, typically for $0.4 < \mathcal{M} < 1.4$.

at $M = 3$, a scramjet coming on line at $M = 6$, and the rocket resuming for orbital insertion. There are numerous alternatives; which ones are practical is a matter of technological development.

1.2 Representative Aircraft

Attributes of several aircraft types are presented in preparation for the more general study of flight dynamics and control. The principal objectives are to indicate the effect that flight conditions and missions have on aircraft configurations and, conversely, to suggest the latitude for motions and control afforded by existing designs. The following descriptions are approximate.

LIGHT GENERAL AVIATION AIRCRAFT

The *Cirrus SR20* (Fig. 1.1-1) is a four-place aircraft with a 149 kW (200 hp) reciprocating engine. The wingspan is 10.8 m, the length is 7.9 m, and the wing area is 12.6 m². The empty and maximum takeoff masses are 885 and 1,315 kg; hence, the maximum wing loading⁵ is 105 kg/m². The cruising airspeed⁶ of this unpressurized aircraft is 296 km/hr (184 statute miles per hour [mph]) at 75% power and 1,980 m (6,500 ft) altitude, and its range is 1,480 km (920 mi). The maximum Mach number is 0.24. The SR20's straight wing has a moderate aspect ratio⁷ (9.1) for efficient subsonic cruising, and it is mounted low on the fuselage with 5 deg dihedral angle. The horizontal tail is also mounted low, and the vertical tail is swept, with a full-length rudder whose throw is unrestricted by the horizontal tail location. The principal airframe material is a structural composite, and the structure uses semimonocoque construction (i.e., the skin carries a significant portion of the loads), with surfaces bonded to spars and ribs. A recovery parachute that can bring the entire aircraft safely to the ground is standard equipment.

VARIABLE-STABILITY RESEARCH AIRCRAFT

The *Princeton Variable-Response Research Aircraft* (VRA) is a highly modified single-engine North American Navion A airplane designed for research on flight dynamics, flying qualities, and control (Fig. 1.2-1). The

5. *Wing loading* is defined in SI units as aircraft mass (kg) divided by wing area (m²). Using U.S. Customary Units, it is defined as weight (lb) divided by wing area (ft²).

6. The *airspeed* is the magnitude of the aircraft's velocity relative to the surrounding atmosphere, i.e., of the *relative wind*. If the wind is blowing, airspeed and ground speed are not the same.

7. The *wing aspect ratio* is defined as the span squared divided by the wing area.



Fig. 1.2-1. Princeton variable-response research aircraft. (courtesy of Princeton University)

most distinguishing feature of the VRA is the pair of vertical side-force-generating surfaces mounted midway between wing roots and tips, but equally important elements of its design are the digital fly-by-wire (DFBW) control system, first installed in 1978, which parallels the standard Navion's mechanical control system, and the fast-acting wing flaps that produce negative as well as positive lift. In operation, a safety pilot/test conductor has direct control of the aircraft through the mechanical system, while the test subject controls the aircraft through the experimental electronic system.

Although limited to an airspeed of about 200 km/hr (125 mph), the VRA can simulate the perturbational motions of other aircraft types through independent, closed-loop control of all the forces and moments acting on the airplane. Feedback control of motion variables to the control surfaces allows the natural frequencies, damping ratios, and time constants of the Navion airframe to be shifted to values representative of other airplanes, while the direct-force surfaces (side force and lift) provide realistic accelerations in the cockpit.

Consider, for example, pure pitching and yawing motions of the 747 airliner. The aircraft rotates about its center of mass, which is considerably aft of the cockpit; consequently, substantial lateral and vertical motions and accelerations are felt in the cockpit when simple angular rotations occur. The VRA can simulate these lateral and vertical motions through



Fig. 1.2-2. Schweizer SGS 2-32 sailplane. (courtesy of Princeton University)

the actions of its direct-force control surfaces. Conversely, these surfaces make it possible to decouple or modify the VRA's natural modes of motion for experimentation. Thus, the VRA can execute a turn without banking or can bank without turning. Its control surfaces can also be used to simulate turbulence for test purposes.

Having been used for over twenty years of research at Princeton University, the VRA and its sister ship, the *Avionics Research Aircraft* (which is virtually identical to the VRA but does not have side-force panels) are currently owned and operated by the University of Tennessee Space Research Institute.

SAILPLANE

The *Schweizer SGS 2-32* sailplane (Fig. 1.2-2) is a three-place aircraft with a wingspan of 17.4 m, an empty mass of 380 kg, and a typical loaded mass of 600 kg. With an aspect ratio of 18.1 and a wing area of 16.7 m²,



Fig. 1.2-3. Cessna Citation CJ2 executive jet transport. (courtesy of Cessna Aircraft Co.)

the aircraft's minimum sink rate is 0.6 m/s. The maximum speed is 255 km/hr, while its speed for maximum lift-drag ratio is about 80 km/hr. The aircraft held numerous speed and distance soaring records in the 1970s, and it was used for aerodynamic studies at Princeton's Flight Research Laboratory. Several 2-32s were purchased by the U.S. government and designated the X-26A. As the QT-2 or X-26B, Lockheed fitted a quiet engine to the aircraft for testing of stealthy reconnaissance concepts that led to the operational YO-3A. The mechanical flight control system is conventional, with all-moving horizontal tail, ailerons, rudder, flaps, and speed brakes (or "spoilers").

BUSINESS JET AIRCRAFT

The *Cessna Citation CJ2* is representative of many small executive jet transports (Fig. 1.2-3); it has a "T" tail, two 10.7 kN (2,400 lb) thrust

turbofan engines in separate nacelles,⁸ and an unswept wing. Mounting the engines on the fuselage aft of the rear pressure bulkhead has several advantages in this aircraft class, including good ground clearance, protection of the engines against ingesting foreign objects while on the ground, reduced cabin noise, low contribution to drag, and structural efficiency. The Model CJ2 carries one or two crew members and up to six passengers in its 14.3-m-long fuselage. The wingspan, area, and aspect ratio are 15.1 m, 24.5 m², and 9.3, while the minimum and maximum masses are 3,460 and 5,580 kg. The maximum wing loading is 228 kg/m². The economical cruise speed at 12,500 m (47,300 ft) altitude is 704 km/hr (437 mph), and the maximum Mach number is 0.72. With maximum fuel, the maximum range of the Model CJ2 is 3,110 km (1,935 mi).

TURBOPROP COMMUTER AIRCRAFT

The *Bombardier Q200* (Fig. 1.2-4) exemplifies the many twin-turboprop designs used by regional airlines to move dozens of passengers on relatively short trips. The Q200 has a “T” tail and a high, straight wing with span, area, and aspect ratio of 25.9 m, 54.4 m², and 12.4, respectively. The craft is 22.3 m long, and it carries 37 passengers plus a crew of two; its empty and maximum masses are 10,380 and 16,465 kg, with a maximum wing loading of 305 kg/m². Typical cruise speed is 535 km/hr (335 mph), corresponding to $M = 0.44$. The maximum operating altitude is at 7,620 m (25,000 ft), and the maximum range is 1,715 km (1,065 mi). The Q200’s primary structure is aluminum, and its engines each generate 1,490 kW (2,000 hp). The aircraft has an active noise and vibration suppression system for quieting the passenger cabin.

SMALL COMMERCIAL TRANSPORT AIRCRAFT

The *Airbus A318* (Fig. 1.2-5), a shortened derivative of the A320, is on the boundary in flight performance and accommodations between typical regional jets and trunkline aircraft. The A320 was the first subsonic commercial jet to use a DFBW control system and sidearm hand controllers in the cockpit, and it was the first to use composite materials for primary structures. Like its predecessor, the A318 has an advanced technology wing, winglets at its wing tips, and a high degree of redundancy in flight control. Hydraulically powered flight control surfaces include spoilers for roll control and speed braking, leading-edge slats for increased lift at higher angles of attack, and conventional surfaces for roll,

8. A *nacelle* is a streamlined fairing over an engine.

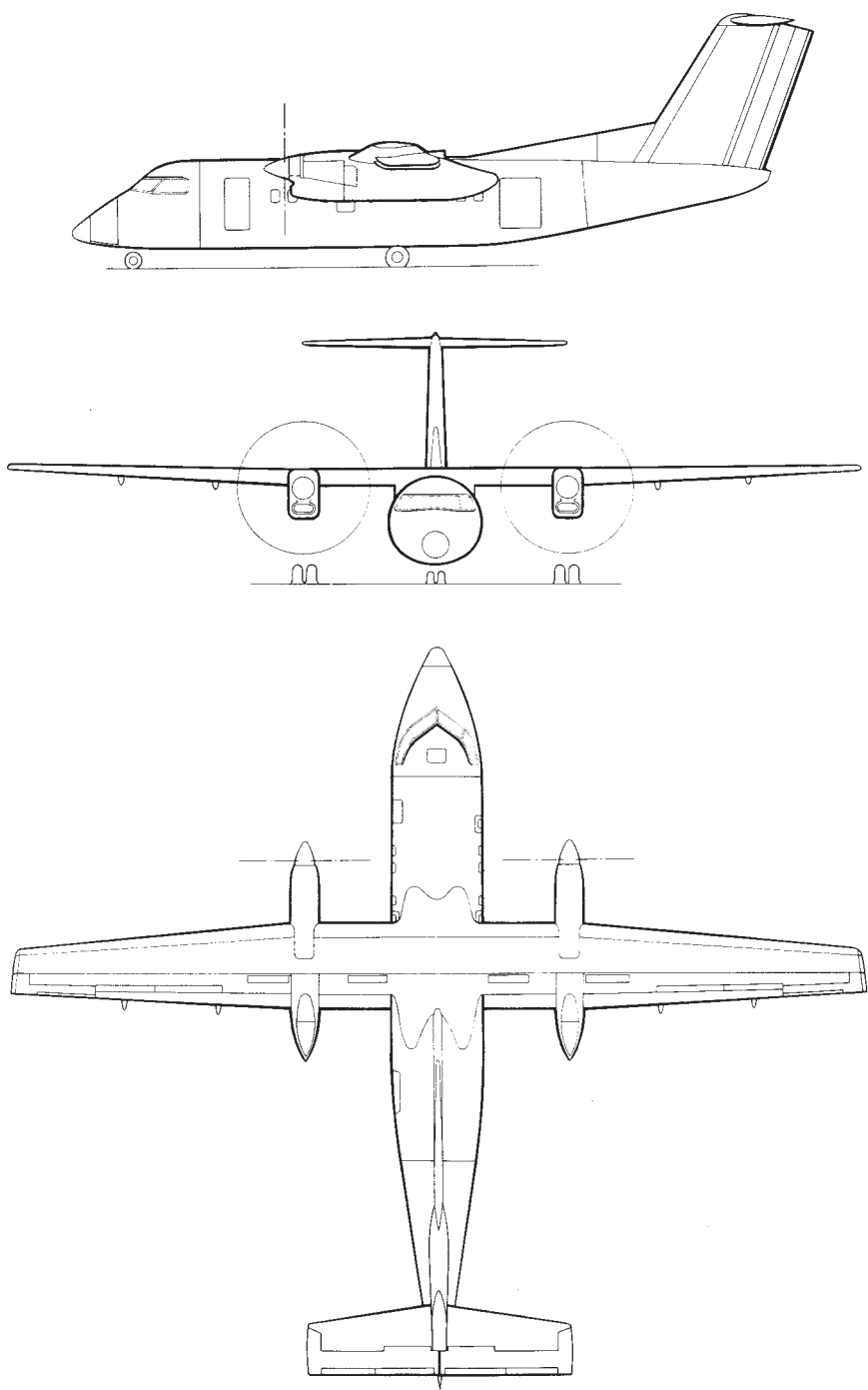


Fig. 1.2-4. Bombardier Q200 regional airliner. (courtesy of Bombardier Aerospace DeHavilland)



Fig. 1.2-5. Airbus A318 small transport aircraft. (courtesy of Airbus Industries)

pitch, and yaw control. There are mechanical backups for rudder control and pitch trim. The single-aisle A318 carries 107–129 passengers, depending on configuration, a flight crew of two, and a cabin crew of three. The empty mass is 85,900 kg, and normal maximum takeoff mass is 59,000 kg. The wingspan and fuselage length are 34.1 and 31.4 m, while the wing area, sweep angle, and aspect ratio are 122 m², 25 deg, and 9.5. Each of the A318's engines produces 98–107 kN (22,000–24,000 lb) of maximum thrust. The optimal cruising Mach number is 0.78. The maximum range with 107 passengers is over 2,750 km.

LARGE COMMERCIAL TRANSPORT AIRCRAFT

At 73.9 m, the *Boeing 777-300* was the longest twin-engine aircraft built at its introduction. It has a maximum takeoff mass of over 299,000 kg and an empty mass of 160,530 kg (Fig. 1.2-6). It is powered by two 435 kN (98,000 lb) thrust turbofan engines mounted in nacelles under the wing. The aircraft has a maximum range of 6,715–15,370 km (3,625–8,300 nautical mi), assuming flight at the economic cruising Mach number of



Fig. 1.2-6. Boeing 777-300 commercial airliner. (courtesy of The Boeing Commercial Aircraft Co.)

0.84, with step climbs in altitude⁹ as fuel is expended. The maximum certified altitude is 13,135 m (43,100 ft). The 777-300 has a wingspan of 60.9 m and its wing area is 427.8 m², giving an aspect ratio of 8.7 and a maximum wing loading of 700 kg/m². The span of the horizontal tail is about double the wingspan of the Cirrus SR20. The double-aisle 777-300 can be configured to carry 450 passengers in two-class seating or 550 passengers in single-class seating.

In spite of its size, the 777 is certified to be flown by a two-person crew. The wing, mounted low on the fuselage, is swept 31.5 deg. Like other large aircraft with long, flexible wings, the 777 has two sets of roll

9. Current air traffic regulations typically require aircraft to cruise at constant altitude, allowing discrete altitude changes between assigned flight levels. With improving navigation and air traffic control systems, continuous cruise-climb will become more common, as it can provide additional fuel savings.



Fig. 1.2-7. Aérospatiale/British Aerospace Concorde commercial airliner. (courtesy of NewsCast Limited)

controls: ailerons near the wing tips for low-speed roll control and mid-wing flaperons plus spoilers for high-speed roll control with reduced wing bending moments. The 777 is the first U.S. commercial transport to use DFBW controls, with a triply redundant system; each of these is internally triply redundant, with dissimilar microprocessors to minimize the impact of hardware or software failures. While control signaling is electronic, the control surface actuators are powered hydraulically.

SUPERSONIC TRANSPORT AIRCRAFT

The *Aérospatiale/British Aerospace Concorde* was the world's only operational supersonic transport aircraft (Fig. 1.2-7), flying from 1976 until 2003. It carried 128 passengers and a crew of three at a maximum Mach number of 2.2 and a typical cruising speed of 2,180 km/hr (1,350 mph). The cruising altitude was over 15,000 m (50,000 ft). Concorde also had an economical subsonic cruising point at $M = 0.93$. The supersonic cruising range with a 10,100 kg payload and reserve fuel was 6,300 km (3,900 mi). Concorde was powered by four turbojet engines, each generating 169 kN (38,050 lb) of thrust with 17 percent afterburning for cruise; full afterburning was available for acceleration through the transonic region, where drag is especially high. The aircraft was 61.7 m long and had a wingspan of 25.6 m; its wing area and aspect ratio were 358.3 m² and 1.82, respectively. With empty and maximum-takeoff masses of 79,265 and 181,435 kg, its average wing loading was 365 kg/m².



Fig. 1.2-8. Lockheed-Martin F-35 joint strike fighter aircraft. (courtesy of Lockheed-Martin Corp.)

Concorde's delta wing had a sweep angle that varied from root to tip in an "ogee" curve. This shape is important both for supersonic cruising efficiency and for satisfactory placement of the strong vortices that form over the wing at high angle of attack during takeoff and landing. The need for good pilot visibility during landing and sharp-nosed streamlining in supersonic flight conflict; hence, the aircraft had a two-position fairing ("droop snoot") in front of the cockpit that accommodated both demands. As there was no separate fore or aft control surface, pitch and roll control were both supplied by the same set of "elevons" (deflected in sum for pitch and in difference for roll) at the wing's trailing edge. The control moment arms were relatively short, and without a separate surface for pitch control, there could be no flaps for landing. Fuel was transferred fore and aft during flight to maintain proper balance as the aerodynamic center shifted with Mach number. Concorde was the first commercial airliner to incorporate a dual-redundant, analog fly-by-wire (FBW) control system in parallel with a mechanical control system.

FIGHTER/ATTACK AIRCRAFT

The *Lockheed-Martin Joint Strike Fighter* is a single-engine, single-seat, high-performance aircraft intended for ground attack and air-to-air combat (Fig. 1.2-8). It is designed for low observability ("stealth") as well as high performance. Three variants with 80–90 percent commonality are planned: a multirole conventional aircraft (F-35A), a carrier-based version (F-35B), and a short-takeoff and vertical-landing (STOVL) version (F-35C). The main turbofan engine for all variants produces an unaugmented

maximum thrust of about 155 kN (35,000 lb), while the lift fan for the F-35C, driven by a shaft from the main engine, produces 80 kN (18,000 lb) of thrust for STOVL operations. The F-35A is 15.5 m long and it has a 10.7 m wingspan. The empty mass is 10,000 kg, while the maximum-takeoff mass is 22,700 kg. Given its wing area of 42.7 m², the aspect ratio is 2.7, and the maximum wing loading is 530 kg/m². The F-35's maximum Mach number is about 1.5, although its typical maneuvering speed would be in the transonic range below Mach 1. The F-35B has a larger wing that folds for storage on an aircraft carrier, as well as larger stabilizing and control surfaces, strengthened landing gear, and a tail hook for arrested landing. The dimensions of the F-35C are similar to those of the F-35A, although the lift-fan engine and attendant structure increase the empty weight. Distinguishing features include twin vertical tails, side-mounted air inlets, and trapezoidal wing and horizontal tail.

BOMBER AIRCRAFT

The *Northrop Grumman B-2 Spirit* (Fig. 1.2-9) is a high-subsonic flying wing configuration with an unrefueled range of about 11,500 km and a payload of 18,000 kg. The angular planform, with straight edges and sharp corners, and the lack of a vertical tail are beneficial characteristics for low observability, although the latter provides a unique flight control challenge: providing adequate yaw stability and control. An active system using deflection of outboard “drag rudders” achieves this goal. There are four control surfaces on each wing, with elevator, aileron, and rudder functions achieved by blending commands to these surfaces. The aircraft's wingspan, wing area, and length are 52.4 m, 465 m², and 21 m. The B-2 is powered by four 77 kN (17,300 lb) turbofan engines.

SPACE SHUTTLE

The *Rockwell Space Shuttle* (Fig. 1.2-10) is the second glider on our list, although it is powered into orbit by dual solid-fuel rockets and its three liquid-oxygen/hydrogen engines. The shuttle also has two orbital maneuvering rockets, as well as small thrusters for attitude control in space. During its return from orbit, the shuttle transitions from spacecraft to aircraft, reentering the atmosphere at $M = 25$ and a high angle of attack to dissipate heat efficiently without damaging the structure, and performing a “dead-stick” horizontal landing at 335 km/hr (210 mph). The shuttle's planform and aerodynamic control surfaces are functionally similar to the Concorde's;



Fig. 1.2-9. Northrop Grumman B-2 Spirit bomber aircraft. (courtesy of Northrop Grumman Corp.)

however, the shuttle has a good deal more drag and thus glides at a much steeper flight path angle. The craft is controlled by a quintuply redundant DFBW system with no mechanical backup. The shuttle carries a flight crew of two plus up to five mission specialists. Its empty mass is 70,600 kg, and it can return to earth with a 14,500 kg payload. With a wing area



Fig. 1.2-10. Rockwell space shuttle. (courtesy of NASA)

of 250 m^2 , the average wing loading during entry is 310 kg/m^2 . The wingspan is 23.8 m , for an aspect ratio of 2.3 , and the vehicle length is 37.2 m .

UNINHABITED AIR VEHICLE

Many missions, including meteorological sampling, communications relay, surveillance, reconnaissance, target acquisition, and ground attack, may not require the presence of a human pilot on board the aircraft, may subject the pilot to high risk, or may require endurance or stress that exceeds normal human capabilities. Uninhabited air vehicles (UAV) are low-cost alternatives for performing these missions, and a number of designs have proved their worth to date. The *General Atomics RQ-1 Predator A* (Fig. 1.2-11) has physical and flight characteristics that are similar to those of a light general aviation aircraft, with a wingspan of 14.8 m , a length of 8.2 m , and maximum mass of $1,020 \text{ kg}$. Powered by a 75 kW (100 hp) reciprocating engine, the RQ-1 has a cruising speed of 130 km/hr , a top speed of 210 km/hr , and a maximum endurance of 40 hr . The Predator A carries a payload of 205 kg , and it has been outfitted to launch



Fig. 1.2-11. General Atomics RQ-1 Predator A uninhabited air vehicle. (courtesy of General Atomics International)

small missiles as well as conduct surveillance. The aircraft has a straight wing, inverted-V tail, and retractable tricycle landing gear. Its ceiling is 7,600 m.

1.3 The Mechanics of Flight

Aircraft flight is described by the principles of classical mechanics. *Mechanics* deals with the motions of objects that possess a substantive scalar inertial property called *mass*. Objects may be modeled as individual particles (also called *point masses*) or assemblages of particles called *bodies*. The translational motions of both point masses and bodies are of interest, as such objects occupy positions in space and may possess three linear velocity components relative to some frame of reference. The three scalar velocity components can be combined in a single three-dimensional column vector $\mathbf{v}$, where

$$\mathbf{v} \triangleq \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad (1.3-1)$$

The product of an object's mass and velocity is called *translational momentum*, also a three-dimensional vector. *Forces* may act on an object to change its translational momentum, which otherwise remains constant relative to an inertial (absolute) frame of reference.

Unlike point masses, bodies have three-dimensional shape and volume; the *position* of a body is defined by the coordinates of a particular reference point on or in the body, such as its *center of mass* (the “balance point”). The velocity of the body refers to the velocity of that reference point. The angular orientation and rotational motion of a body are also important descriptors of its physical state. *Angular momentum*, the rotational equivalent of translational momentum, remains unchanged unless a *torque* (a force that is displaced from the center of mass by a moment arm and that is perpendicular to the moment arm) acts on the body. A body can be characterized by six rotational inertial properties called *moments* and *products of inertia*; the former portray the direct relationship between angular rate and momentum about a rotational axis, while the latter establish coupling effects between axes.

Mechanics is further broken down into kinematics, statics, dynamics, and control. *Kinematics* is the general description of an object's motions without regard to the forces or torques that may induce change; thus, the geometry and coupling of position and velocity (both translational and rotational) are considered, while the means of effecting change are not. *Statics* addresses the balance of forces and torques with inertial effects to produce equilibrium. An aircraft can achieve static equilibrium when it is moving, as long as neither translational nor angular momentum is changing; for constant mass and rotational inertial characteristics, this implies unaccelerated flight. *Dynamics* deals with accelerated flight, when momentum is changing with time. While it is possible to achieve a steady, dynamic equilibrium (as in constant-speed turning flight), the more usual dynamics problem concerns continually varying motions in response to a variety of conditions, such as nonequilibrium initial conditions, disturbance inputs, or commanded forces and torques.

We refer to linear and angular positions and rates of the aircraft as its dynamic *state*; the corresponding twelve quantities are arrayed in the *state vector* described below. Motions occurring in the vertical plane are called *longitudinal motions*, while those occurring out of the plane are called *lateral-directional motions*. The variables of longitudinal motion, related to body axes, are axial velocity, normal velocity, pitching rate, and their inertial-axis integrals: range, altitude, and pitch angle. The lateral-directional variables are side velocity, roll rate, yaw rate, and their inertial-axis integrals, crossrange, roll angle, and yaw angle. *Stability* is an important dynamic characteristic that describes the tendency for the aircraft's state to return to an equilibrium condition or to diverge in response to inputs or initial conditions. *Control* is the critical area of

mechanics that develops strategies and systems for achieving goals and assuring stability, given an aircraft's mission, likely disturbances, parametric uncertainties, and piloting tasks. The aircraft's natural stability can be augmented by *feedback control*, in which measured flight motions drive control effectors on a continuing basis.

For the most part, *kinematics* is well described by algebraic and trigonometric equations. Because many scalar variables often must be addressed simultaneously, matrix-vector notation is helpful for kinematic and other problems. For example, suppose $\mathbf{x}$ is a vector with three components (x_1, x_2, x_3) and $\mathbf{y}$ is a vector with two components (y_1, y_2) such that

$$y_1 = x_1 + \sin x_2 + \cos x_3 \quad (1.3-2)$$

$$y_2 = \cos x_1 + \sin x_3 \quad (1.3-3)$$

The two equations can be replaced symbolically by the single vector equation

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) \quad (1.3-4)$$

where $\mathbf{f}(\mathbf{x})$ is a two-component vector composed of the right sides of eqs. 1.3-2 and 1.3-3. As a second example, let

$$y_1 = ax_1 + bx_2 + cx_3 \quad (1.3-5)$$

$$y_2 = dx_1 + ex_2 + fx_3 \quad (1.3-6)$$

The two scalar equations are represented by the matrix-vector equation

$$\mathbf{y} \triangleq \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{H}\mathbf{x} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad (1.3-7)$$

where $\mathbf{H}$ is the (2×3) *matrix*

$$\mathbf{H} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \quad (1.3-8)$$

and the matrix-vector product is defined accordingly.¹⁰

The scalar velocity v is the time rate of change (or derivative with respect to time) of a scalar position r ,

$$v = dr/dt \quad (1.3-9)$$

10. Generic scalar quantities are denoted by lower-case italic letters (a, b, c), while vectors are represented by lower-case bold letters ($\mathbf{a}, \mathbf{b}, \mathbf{c}$) and matrices by upper-case bold letters ($\mathbf{A}, \mathbf{B}, \mathbf{C}$). Italic letters (upper or lower case) and other fonts are used throughout to express commonly accepted symbols without rigid adherence to this convention.

while the position is the time integral of the velocity plus the appropriate constant of integration:

$$r(t) = r_0 + \int_0^t v(t) dt \quad (1.3-10)$$

The equations of *statics* can be combined in a single vector equation

$$0 = f(\mathbf{x}, \mathbf{u}, \mathbf{w}) \quad (1.3-11)$$

which indicates that some combination of forces and inertial effects is simultaneously balanced along many coordinate directions. In eq. 1.3-11, $f(\cdot)$ represents the vector of scalar momentum equations to be balanced, $\mathbf{x}$ is the solution variable (e.g., a six-component reduced state vector of translational and rotational velocities), $\mathbf{u}$ is a vector of constant control displacements, and $\mathbf{w}$ is a vector of constant disturbances.

In the *dynamic* case, the physical equations may vary in time t , and rather than equaling zero, they equal the rates of change of the motion variables, term by term:

$$\frac{d\mathbf{x}(t)}{dt} \triangleq \dot{\mathbf{x}}(t) = \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t), \mathbf{w}(t), t] \quad (1.3-12)$$

This is an *ordinary differential equation* for the *state vector* $\mathbf{x}(t)$ as a function of the independent variable t . The definitions of $\mathbf{x}$ and $\mathbf{f}[\cdot]$ can be extended to include position variables and their kinematics. A first-order differential equation of this form is called a *state-space model* of system dynamics.

The object of control is to apply forces and moments that produce desired values of the state during the time interval of interest (t_0, t_f). The control vector $\mathbf{u}(t)$ may be a function of any measurable variables, of time, or of external inputs such as pilot commands $\mathbf{u}_c(t)$; hence, the dynamic equation becomes

$$\dot{\mathbf{x}}(t) = \mathbf{f}[\mathbf{x}(t), \mathbf{u}[\mathbf{x}(t), \mathbf{w}(t), \mathbf{u}_c(t), t], \mathbf{w}(t), t] \quad (1.3-13)$$

in the controlled case.

It is rarely adequate to know just the form of the aircraft's differential equation: we want to know the state as a function of time or some other variable. Therefore, equation 1.3-13 must be integrated subject to the initial condition of the state $\mathbf{x}(t_0)$:

$$\mathbf{x}(t) = \mathbf{x}(t_0) + \int_{t_0}^t \mathbf{f}[\mathbf{x}(t), \mathbf{u}[\mathbf{x}(t), \mathbf{w}(t), \mathbf{u}_c(t), t], \mathbf{w}(t), t] dt \quad (1.3-14)$$

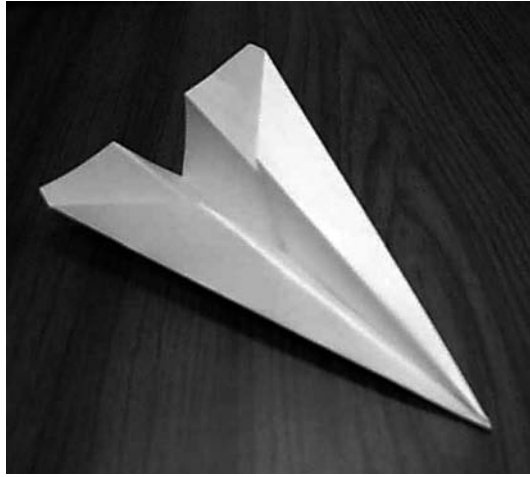


Fig. 1.3-1. Classic paper airplane.

It is always possible to integrate eq. 1.3-13 numerically; however, simpler alternatives may provide greater insight into aircraft behavior. If, for example, $f\{\cdot\}$ is a linear function of its arguments, the *initial condition response* can be written as the matrix-vector product

$$\mathbf{x}(t) = \Phi(t, t_0) \mathbf{x}(t_0) \quad (1.3-15)$$

where $\Phi(t, t_0)$ is a *state transition matrix* relating the state at time t_0 to the state at time t . It is also possible to write the steady sinusoidal response of a linear system to a steady sinusoidal control input $\mathbf{u}(t) = \mathbf{a} \sin \omega t$ with vector amplitude $\mathbf{a}$ at frequency ω as

$$\mathbf{x}(\omega) = \mathbf{A}(j\omega) \mathbf{u}(\omega) \quad (1.3-16)$$

where $\mathbf{A}(j\omega)$ is the system's complex-valued *frequency-response matrix* and $j \triangleq \sqrt{-1}$. Replacing $j\omega$ by the complex operator, $s \triangleq \sigma + j\omega$, $\mathbf{A}(s)$ is found to be the *transfer function matrix*, an important element of stability and control analysis.

Example 1.3-1. Longitudinal Motions of a Paper Airplane

The classic dart-shaped paper airplane (figure 1.3-1) provides an excellent introduction to flight mechanics, for both numerical analysis and experiment. Starting with a sheet of plain paper, which weighs¹¹ about 3 g,

11. Strictly speaking, its *mass* (not weight) is 3 g; “weighs” is used here and throughout in a colloquial sense.

the experimental vehicle has a wingspan of 12 cm and a length of 28 cm, yielding an aspect ratio AR and wing area S of 0.86 and 0.017 m^2 , respectively. As shown in Chapter 3, the point-mass longitudinal motions of the paper airplane can be predicted by integrating four differential equations,

$$\dot{V} = -C_D \left(\frac{1}{2} \rho V^2 \right) S / m - g \sin \gamma$$

$$\dot{\gamma} = \left[C_L \left(\frac{1}{2} \rho V^2 \right) S / m - g \cos \gamma \right] / V$$

$$\dot{h} = V \sin \gamma$$

$$\dot{r} = V \cos \gamma$$

where V , γ , h , and r are the airspeed, flight path angle, height, and range, respectively. The parameters of this model are contained in the drag coefficient C_D , lift coefficient C_L , air density ρ (1.225 kg/m^3 at sea level), reference area S , mass m , and acceleration due to gravity, $g = 9.807 \text{ m/s}^2$. The lift coefficient is modeled as a linear function of the angle of attack α ,

$$C_L = C_{L_\alpha} \alpha$$

The lift-slope derivative is estimated as

$$C_{L_\alpha} = \frac{\pi AR}{1 + \sqrt{1 + (AR/2)^2}} \quad (\text{per radian})$$

The drag coefficient is modeled as

$$C_D = C_{D_0} + \epsilon C_L^2$$

with $C_{D_0} = 0.02$ and the *induced-drag factor* $\epsilon = 1/\pi e AR$; e is known as the *Oswald efficiency factor*, estimated to be 0.9 here. In this example, attitude dynamics are neglected, so the angle of attack is taken as the control variable; its trimmed value is a function of the center-of-mass location and the amount of upward deflection of the wing's trailing edge.

This mathematical model can be put in the form of eq. 1.3-12 by making the following definitions:

$$\mathbf{x} \triangleq \begin{bmatrix} V \\ \gamma \\ h \\ r \end{bmatrix} \triangleq \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \mathbf{u} \triangleq \alpha \triangleq u$$

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) \triangleq \begin{bmatrix} -C_D(u) \left(\frac{1}{2} \rho x_1^2 \right) S/m - g \sin x_2 \\ \left[C_L(u) \left(\frac{1}{2} \rho x_1^2 \right) S/m - g \cos x_2 \right] / x_1 \\ x_1 \sin x_2 \\ x_1 \cos x_2 \end{bmatrix}$$

$\mathbf{f}(\mathbf{x}, \mathbf{u})$ is a nonlinear function of its arguments for several reasons: $C_D(u)$ is a quadratic function of u , and several nonlinear operations involve elements of the state, including the square of x_1 , division by x_1 , sines and cosines of x_2 , and the product of x_1 with functions of x_2 .

Three longitudinal paths are typical of paper airplane flight: *constant-angle descent*, *vertical oscillation*, and *loop*. Numerically integrating the equations of our mathematical model produces these paths (eq. 1.3-14), as shown in the accompanying figures. Four cases are simulated: (a) equilibrium glide at maximum lift-drag ratio (L/D), (b) oscillating glide due to zero initial flight path angle, (c) increased oscillation amplitude due to increased initial speed, and (d) loop due to a further increase in launch speed. In all cases, α is fixed at 9.3 deg, yielding the best L/D (5.2) for the configuration. This angle of attack produces the shallowest equilibrium glide slope (−11 deg) and a constant speed of 3.7 m/s. The equilibrium trajectory is represented by the straight descending path in Fig. 1.3-2a and the constant speed and flight path angle in Fig. 1.3-2b (case a).

Obtaining this smooth, linear path is dependent on starting out with exactly the right speed and angle; case b illustrates that having the right speed but the wrong initial angle (0 deg) perturbs the dynamic system, introducing a lightly damped oscillation about the equilibrium path. The period of the oscillation is 1.7 s, and the wave shape is very nearly sinusoidal.

The initial-condition perturbation is increased in case c, where the zero initial angle is retained, and the speed is increased. The vertical oscillation has greater height, speed, and angle amplitude, and the wave shape is perceptibly scalloped in the first two variables. Nevertheless, the rate of amplitude decay and the period are about the same as in the previous case. Further increasing the launch speed results in a loop, followed by sharply scalloped, decaying undulations with a 1.7 s period (case d). The loop provides an upper limit on the vertical oscillations that can be experienced by an airplane with fixed control settings.

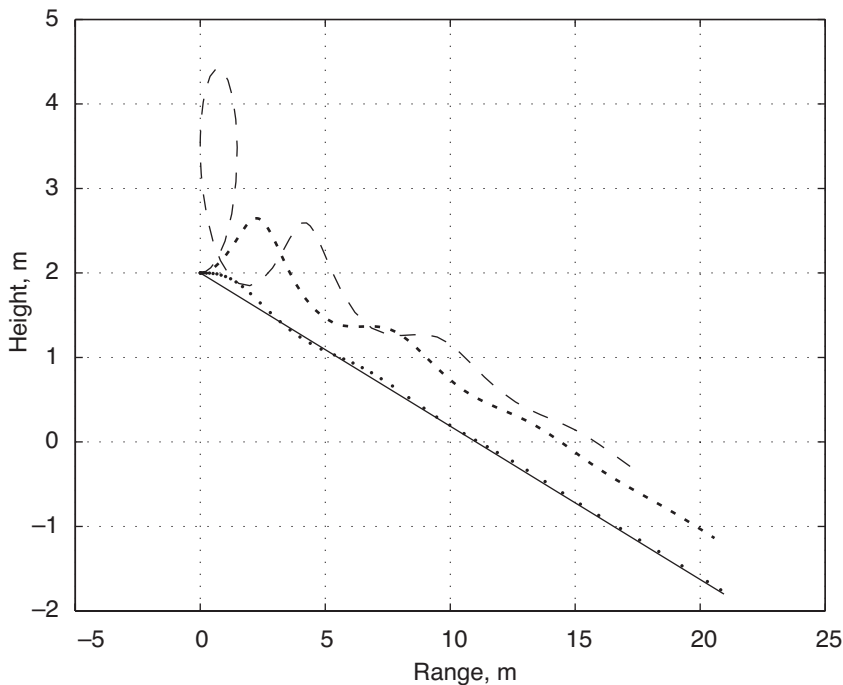


Fig. 1.3-2a. Flight of a paper airplane: Height versus range.

The MATLAB program used to generate these results can be found in Appendix D. You can replicate these flight paths with your own paper airplane, although you are not likely to get identical flight test results. The aerodynamic model used here is approximate, and the weight and dimensions of your airplane will be slightly different. It is difficult to trim the paper plane's α precisely, but it is relatively easy to sweep a range of values by bending the wing's trailing edge up or down. Similarly, launching by hand produces variable starting conditions, and it probably is not possible to launch a paper airplane at speeds much above 5 m/s without inducing significant aeroelastic deformations to the structure, changing its aerodynamics. Perhaps most important, attitude dynamics (not modeled above) play a role, particularly in large-amplitude maneuvers such as the loop. It is tricky to loop the classic dart-shaped paper airplane in practice, but other designs can be looped readily.

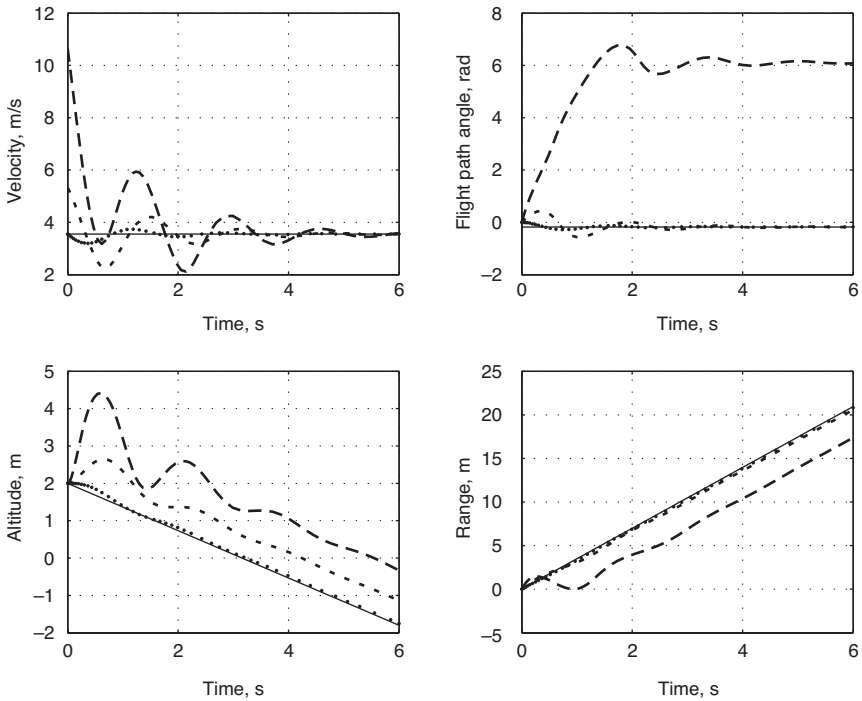


Fig. 1.3-2b. Flight of a paper airplane: Velocity, flight path angle, altitude, and range versus time.

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Chapter 2

Exploring the Flight Envelope

The range of static and dynamic conditions within which an aircraft operates is called its *flight envelope*. To explore the flight envelope, we first must establish frames of reference and define basic elements of position, velocity, force, moment, and equilibrium. Our attention is directed to steady or quasisteady flight, and the aircraft is modeled as a rigid body flying over a flat earth. *Steady flight* is characterized by constant velocity and altitude, with unchanging aircraft mass, as in wings-level cruise, where there is a balance between lift, weight, thrust, and drag. Rigorously, steady flight is possible only if thrust is produced without changing the aircraft's mass. *Quasisteady flight* occurs when time variations in mass or flight condition are slow enough to produce negligible dynamic effects, such as small rate of weight loss due to fuel burn, shallow climbing and gliding, and turns with constant bank angle and yaw rate.

The *rigid-body assumption* allows the aircraft's translational position and angular attitude to be defined by the position of its center of mass and the orientation of a body-fixed coordinate system relative to an inertial frame of reference. It limits the degrees of freedom needed to describe aircraft motion and eliminates the dependence of aerodynamic and inertial effects on structural bending and torsion. Body flexibility affects structural loads and ride quality; it must be considered in flight control design, but it has little direct effect on the net position and velocity.

The *flat-earth assumption* neglects the variation of gravitational acceleration with distance from the center of the earth, and it disregards relatively subtle Coriolis and centrifugal effects. The description of position and velocity relative to the earth is simplified, a useful attribute for investigating the dynamics and control of flight within the atmosphere at suborbital speeds and for studying all modes of motion with periods or time constants less than several hundred seconds. *Round-earth effects* are important for analyzing navigation and guidance, especially for long-distance flights and for trajectories that enter the atmosphere from orbit.

Whatever the frame of reference, we assume in the remainder of the book that flight motions are described by *Newton's laws of motion*, which can be summarized as follows:

First law. If no force acts on a particle, it remains at rest or continues to move in a straight line at constant velocity, as measured in an inertial reference frame.

Second law. A particle of fixed mass acted upon by a force moves with an acceleration proportional to and in the direction of the force; the ratio of force to acceleration is the mass of the particle.

Third law. For every action, there is an equal and opposite reaction.

In this chapter, we focus on steady or quasisteady flight in various conditions, such as climbing, cruising, descending, and turning. We begin with a discussion of the atmosphere and its characteristics, noting the changes of air pressure, density, and temperature with altitude, as well as descriptions of the wind field (Section 2.1). We examine frames of reference and the relationships between them. Euler angles portray the orientation of one frame with respect to another; hence, transformations between reference frames must involve these angles (Section 2.2). The static force, moment, and inertial properties establish equilibrium conditions that are important for steady flight (Section 2.3). The corresponding aerodynamic effects and nondimensional coefficients commonly used to express aerodynamic properties and the generation of thrust by powerplants are presented (Sections 2.4 and 2.5). The chapter concludes with a review of the principles that govern aircraft performance (Section 2.6).

2.1 The Earth's Atmosphere

The actual values of the air's pressure, density, and temperature vary with the seasons and depend upon the weather and time of day, but standard atmospheric conditions can be used in many flight calculations. A brief summary of the U.S. Standard Atmosphere, 1976 [A-7] is presented in Fig. 2.1-1 and Table 2.1-1. A MATLAB function (`Atmos`) for interpolating air density, air pressure, temperature, and the speed of sound from the standard atmosphere table can be found in Appendix B. The systematic variability of air density from these average values is less than 1 percent below 10 km altitude and less than 20 percent below 50 km. Annual absolute temperature variations of 10 percent in this altitude region are typical. In the lower atmosphere, where aircraft normally fly, the mean free path of the air's molecules is short compared to aircraft dimensions.¹

1. The *Knudsen number* equals the mean free path divided by some characteristic length of the vehicle. The mean free path of the air's constituent molecules is about 0.1 m at an altitude of 100 km.

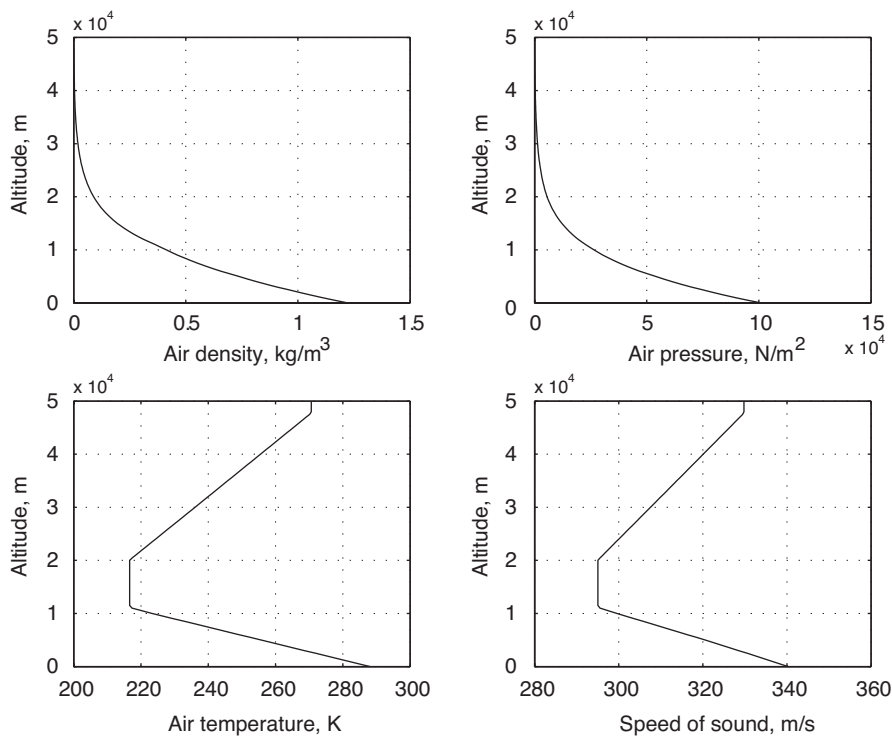


Fig. 2.1-1. Air density, pressure, temperature, and speed of sound at altitudes from 0 to 50 km.

TABLE 2.1-1 U.S. STANDARD ATMOSPHERE, 1976 [A-7]

Z , m	H , m	p/p_o^*	ρ/ρ_o^{**}	$\mathcal{T}$, K	a , m/s	v/v_o^{***}
0	0	1.000	1.000	288.150	340.294	1.000
2,500	2,499	0.737	0.781	271.906	330.563	1.223
5,000	4,996	0.533	0.601	255.676	320.545	1.514
10,000	9,984	0.262	0.338	223.252	299.532	2.413
11,100	11,081	0.221	0.293	216.650	295.069	2.709
15,000	14,965	0.120	0.159	216.650	295.069	4.997
20,000	19,937	0.055	0.073	216.650	295.069	10.946
47,400	47,049	0.0011	0.0011	270.650	329.799	822.11
51,000	50,594	0.0007	0.0007	270.650	329.799	1,286.

$^*p_o = 1.013 \times 10^5 \text{ N/m}^2$ $^{**}\rho_o = 1.225 \text{ kg/m}^3$ $^{***}v_o = 1.461 \times 10^{-5} \text{ m}^2/\text{s}$

Therefore, it is generally acceptable to characterize the atmosphere as a gaseous continuum when studying aircraft flight.

PRESSURE, DENSITY, AND THE SPEED OF SOUND

Atmospheric conditions are related by an “equation of state” and are governed by hydrostatic forces arising from the earth’s gravitational field. The *thermal equation of state* for a perfect gas is

$$p = \rho \left(\frac{R^*}{M} \right) \mathcal{T} = \rho R \mathcal{T} \quad (2.1-1)$$

where p is the *ambient pressure* (pascals, or N/m²), R^* is the *universal gas constant* (8.314 J/mol-K),² M is the *mean molecular weight of air* (28.96 g/mol), R is the *gas constant of air* (287.05 N-m/kg-K), and $\mathcal{T}$ is the *absolute temperature* (K). The *hydrostatic equation* relates differential change in ambient pressure dp to a unit weight of air ρg_{eff} times the differential change in altitude dZ (positive up):

$$dp = -\rho g_{\text{eff}} dZ \quad (2.1-2)$$

The pressure at the earth’s surface is the integral of eq. 2.1-2, that is, the weight per unit area of a column of air that extends from the surface to infinite height.

The effective gravity g_{eff} is a function of altitude. This is a consequence of the earth’s mass attraction plus the centrifugal effect of the earth’s rotation (assuming that the atmosphere rotates at the same rate), both of which can be described as potential functions (Section 3.1). Thus, a distinction is made between the *geometric altitude* Z , referenced to the earth’s surface, and the *geopotential altitude* $H(Z)$, which is defined as

$$H(Z) \triangleq \int_0^Z \left(\frac{g(z)}{g_o} dz \right) = \frac{1}{g_o} [\mathcal{V}(Z) - \mathcal{V}(0)] = \frac{r_m Z}{r_m + Z} \quad (2.1-3)$$

g_o is the net gravitational acceleration at sea level, and $\mathcal{V}(\cdot)$ is the net *potential energy*, described in Chapter 3 and succeeding chapters. For a spherical earth with inverse-square gravitational field whose mean radius r_m is 6,367,474 m, the relationship between H and Z takes the simple form shown. The distinction between H and Z is not great in the lower atmosphere, as can be seen from Table 2.1-1.

The *U.S. Standard Atmosphere* characterizes the temperature by segments that are either constant with altitude or linear in $H(Z)$, as derived from experimental data. The speed of sound a (m/s) is related to the

2. J stands for *Joule*, a unit of energy that is identical to a Newton-meter. It is the work done by one Newton of force moving one meter (see Chapter 3).

absolute temperature by

$$a = \sqrt{\gamma R \mathcal{T}} \quad (2.1-4)$$

where γ , the *ratio of the specific heats* of air at constant pressure and volume (c_p/c_v), is 1.4. The *standard pressure* is modeled as a function of $H(Z)$. In *constant-temperature layers*, the function is

$$p = p_B e^{-g_o(H - H_B)/R\mathcal{T}} = p_B e^{-\beta(H - H_B)} \quad (2.1-5)$$

with $(\cdot)_B$ representing the value at the base of the layer. The quantity $\beta (= g_o/R\mathcal{T})$ is called the *inverse scale height* of the pressure function; it has units of m^{-1} . In a *linearly varying temperature layer*, the standard pressure is given by

$$p = p_B \left[\frac{\mathcal{T}_B}{\mathcal{T}_B + k(H - H_B)} \right]^{g_o/Rk} \quad (2.1-6)$$

where k is the slope of the temperature variation, $\partial \mathcal{T} / \partial H$. The density can then be obtained from the pressure using the equation of state (eq. 2.1-1).

The *dynamic pressure* $\bar{q}$ is a function of flight conditions; it can be expressed in terms of air density ρ and true airspeed V_A or of ambient atmospheric pressure p and Mach number ($\mathcal{M} = V_A/a$). From the above equations,

$$\bar{q} \triangleq \frac{1}{2} \rho V_A^2 = \frac{\gamma}{2} p \mathcal{M}^2 \quad (2.1-7)$$

It is often useful to define an exponential model of air density for trajectory calculations within a given geometric altitude band (z_1, z_2), with positive altitude corresponding to negative z :

$$\rho(z) = \rho_1 e^{\beta(z - z_1)} \quad (2.1-8)$$

If the temperature is constant in the altitude band, β is defined by eq. 2.1-5. If the air temperature varies in (z_1, z_2), a least-squares curve fit to the tabulated data could be used to estimate the equivalent β . A simple expedient is to read ρ_1 and ρ_2 from the U.S. Standard Atmosphere and then compute

$$\beta = \frac{\ln(\rho_2/\rho_1)}{(z_2 - z_1)} \quad (2.1-9)$$

for the interval, as in the Atmos function of Appendix B.

VISCOSITY, HUMIDITY, AND RAIN

Other atmospheric characteristics with flight-dynamic implications include kinematic viscosity, humidity, precipitation, icing, and wind [D-1].

The *kinematic viscosity* ν , which is the ratio of dynamic viscosity to air density, characterizes the shear stress property of air (per unit mass). Large values encourage layered, sliding (*laminar*) flow between adjacent air-streams of differing velocity. At 100 percent relative *humidity* and 300 K, the water vapor concentration in the air can reach about 25 g/m³. Given its lower molecular weight, the water vapor reduces the dry atmosphere's density by about 1 percent in this case.

Although relatively infrequent, precipitation rates of more than 1 m/hr have been measured at the earth's surface. Heavy *rain* can reduce aircraft performance by increasing effective drag, reducing maximum available lift, and adding weight to the aircraft [A-10, T-2, V-3]. In at least one case, rain is believed to have quenched a transport aircraft's jet engines. The effect is transient and presents an immediate hazard to most aircraft only when they are near the ground, operating at high angle of attack and power setting. Rain destroys the laminar flow on which many sail-planes depend for high performance, leading to pitch trim changes as well as lift and drag effects. *Icing*, which occurs when an aircraft flies into supercooled water droplets, is even more insidious, as the ice accumulates on the plane's skin, spoiling airflow, increasing weight, and possibly jamming control surfaces.

WIND FIELDS AND ATMOSPHERIC TURBULENCE

Wind is the most prevalent and complex natural disturbance to aircraft flight. Changing wind fields have a number of meteorological origins. They can be characterized as gust fronts, sea-breeze fronts, air-mass fronts, gravity or solitary waves, terrain-induced wind shear, mountain waves (including wind rotors), low- and high-altitude jet streams, tornadoes, microbursts, and turbulence [T-4]. Flying aircraft leave a trail of intense, localized vorticity in their wakes that poses a hazard to other aircraft. Wind models produce force and moment inputs through the aircraft's aerodynamics; their vector-matrix descriptions are subject to the same matrix transformations used throughout this chapter.

The wind is a vector field that varies in space and time; its magnitude may approach the aircraft's normal airspeed during takeoff and landing. Furthermore, the change in the wind—the *wind shear*—can have persistent effects on aeroelastic modes, as well as dramatic and disastrous effects on the flight path when flying near the ground [T-4]. If the wind is modeled as

$$\mathbf{w}(\mathbf{r}, t) = \begin{bmatrix} w_x(\mathbf{r}, t) \\ w_y(\mathbf{r}, t) \\ w_z(\mathbf{r}, t) \end{bmatrix} \quad (2.1-10)$$

then the *spatial wind shear* must be represented as the *Jacobian matrix* of $\mathbf{w}$ with respect to the position $\mathbf{r}$,

$$\mathbf{w}_r \triangleq \mathbf{W} \triangleq \begin{bmatrix} \frac{\partial w_x(\mathbf{r}, t)}{\partial x} & \frac{\partial w_x(\mathbf{r}, t)}{\partial y} & \frac{\partial w_x(\mathbf{r}, t)}{\partial z} \\ \frac{\partial w_y(\mathbf{r}, t)}{\partial x} & \frac{\partial w_y(\mathbf{r}, t)}{\partial y} & \frac{\partial w_y(\mathbf{r}, t)}{\partial z} \\ \frac{\partial w_z(\mathbf{r}, t)}{\partial x} & \frac{\partial w_z(\mathbf{r}, t)}{\partial y} & \frac{\partial w_z(\mathbf{r}, t)}{\partial z} \end{bmatrix} \quad (2.1-11)$$

while the *temporal wind shear* is the partial time derivative of $\mathbf{w}$:

$$\mathbf{w}_t \triangleq \begin{bmatrix} \frac{\partial w_x(\mathbf{r}, t)}{\partial t} \\ \frac{\partial w_y(\mathbf{r}, t)}{\partial t} \\ \frac{\partial w_z(\mathbf{r}, t)}{\partial t} \end{bmatrix} \quad (2.1-12)$$

The total time derivative of the wind at a point moving through the wind field with velocity $\mathbf{v}_I$ is the sum of the two:

$$\dot{\mathbf{w}} = \mathbf{w}_t + \mathbf{W}\mathbf{v}_I \quad (2.1-13)$$

The varying spatial wind distribution implied by eqs. 2.1-10 and 2.1-11 subjects different parts of an aircraft to different instantaneous winds, further complicating dynamic response. Consider a vertical wind field that has higher upward velocity at the left wing tip than at the right; the airplane is subjected to a positive rolling moment that cannot be computed from the wind velocity at a single point. If the wind distribution is linear across the span, then the rolling effect can be predicted from the stability derivatives presented in Section 3.4. If the distribution is more detailed, then the moment calculation involves a fine-scale integration of spanwise effects.

Isolated *atmospheric vortices*, including tornadoes and wind rotors [W-8], can be modeled as circular flows about a linear axis. The tangential velocity V_T of an inviscous, two-dimensional *ideal vortex* is expressed as

$$V_T = \frac{k}{r} \quad (2.1-14)$$

where k reflects the strength of the vortex and r is the radius from the vortex axis [S-1]. The viscosity limits the velocity as the radius becomes small,

and real vortices are observed to have zero V_T at the axis. The *Lamb-Oseen vortex* imposes an exponential decay on eq. 2.1-14, producing

$$V_T = \frac{k[1 - e^{-(r/r_c)^2}]}{r} \quad (2.1-15)$$

where r_c is a reference radius. The tangential velocity is plotted in Fig. 2.1-2, with $k = r_c = 1$. The maximum value of V_T occurs when $dV_T/dr = 0$; the corresponding radius is $1.12r_c$, and the maximum tangential velocity is $0.64k/r_c$. The flow within the radius r_c approximates a *solid-body rotation* of the air.

A constant-strength, three-dimensional vortex is extended along a line that is perpendicular to the circular motion. Assuming that the axis is aligned with the x direction, the resulting cylindrical flow can be expressed as

$$\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{-k[1 - e^{-(r/r_c)^2}]z}{r^2} \\ \frac{k[1 - e^{-(r/r_c)^2}]y}{r^2} \end{bmatrix} \quad (2.1-16)$$

where y and z are the lateral and vertical distances from the axis. An aircraft leaves a *pair of vortices* of opposing rotational directions in its wake. Their intensity is proportional to the wing's lift and, consequently, to the aircraft's weight. The resulting wind disturbance can be modeled by adding two images of eq. 2.1-16, with r , y , and z referenced to the left and right rotational axes.

Atmospheric turbulence is a random variation that is superimposed on the slowly varying mean wind field:

$$\mathbf{w}(\mathbf{r}, t) = \mathbf{w}_o(\mathbf{r}, t) + \Delta\mathbf{w}(\mathbf{r}, t) \quad (2.1-17)$$

$\mathbf{w}_o(\mathbf{r}, t)$ characterizes a component that measurably persists over some time and region. While the statistical properties of $\Delta\mathbf{w}(\mathbf{r}, t)$ may be relatively constant, the actual wind-speed variations that occur are probabilistic and, therefore, not strictly predictable. $\Delta\mathbf{w}(\mathbf{r}, t)$ is a random process whose components are characterized by probability distributions, expected values, and power spectral densities, which are discussed in Section 4.5.

For the moment, it suffices to describe turbulence as a wavelike phenomenon whose component amplitudes vary with frequency. The von

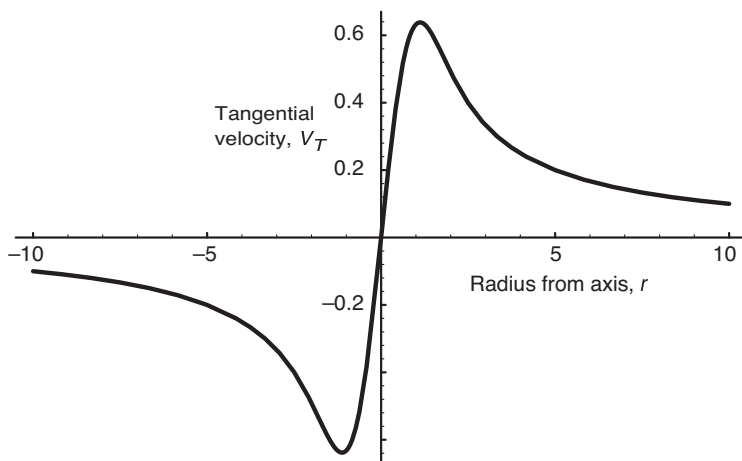


Fig. 2.1-2. Lamb-Oseen vortex.

Kármán turbulence spectra [V-3] are widely accepted representations of atmospheric turbulence in which the incremental wind “power” $\Phi(\Omega)$ (m^2/s^2) (i.e., velocity squared) is a function of wavelength or wave frequency Ω (rad/m). For example, a one-dimensional spectrum for disturbances that are transverse to the flight direction is

$$\Phi(\Omega) = \frac{\sigma^2 L}{2\pi} \frac{[1 + 8/3(aL\Omega)^2]}{[1 + (aL\Omega)^2]^{1/6}} \quad (2.1-18)$$

where σ , a (≈ 1.4), and L ($\approx 1,750$ m) are constants of the random process. Note that the turbulence power decreases with increasing frequency (i.e., decreasing wavelength) as $\Omega^{-11/6}$, close to the $\Omega^{-5/3}$ decrease produced by the Kolmogorov law [L-4].

In later sections, we shall use rational transfer functions and their associated state-space models to describe aircraft response to inputs. The noninteger exponent of the von Kármán spectrum is not easily modeled in state-space format. The Dryden model, though a less accurate fit to experimental data, has no such problem [H-4, H-5]. For transverse turbulence, the Dryden spectrum takes the form

$$\Phi(\Omega) = \frac{\sigma^2 L}{\pi} \frac{[1 + 3(L\Omega)^2]}{[1 + (L\Omega)^2]^2} \quad (2.1-19)$$

while for axial turbulence (i.e., along the line of flight), it is

$$\Phi(\Omega) = \frac{2\sigma^2 L}{\pi} \frac{1}{[1 + (L\Omega)^2]^2} \quad (2.1-20)$$

with $L \approx 1,250$ m.

When significant features of the wind field are large compared to aircraft size, it is satisfactory to consider eqs. 2.1-10 and 2.1-13 as *exogenous* (i.e., external) *disturbances* to the aircraft's dynamics, neglecting the converse effect of the aircraft's presence on the wind field itself. In this case, the aircraft is transparent to the wind; the wind's effects are assumed to be applied at the aircraft's center of mass and across the aircraft's surface. However, for intense small-scale features like the wing-tip vortices generated by another aircraft, the coupling between wind-field dynamics and aircraft dynamics cannot be ignored. Figure 2.1-3 illustrates the impingement of the tip vortex (marked by smoke) generated by a leading Lockheed-Martin C-130 aircraft on a trailing Boeing 737 aircraft. Note that the vortex core, which is essentially straight before reaching the 737, is deflected as it passes over the aircraft's surface.

2.2 Kinematic Equations

Kinematics deals with the geometry of translational and rotational motion. *Position* can be measured with respect to a *basic reference frame* that has three mutually perpendicular directions, customarily denoted by *unit vectors* ($\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$), and corresponding *lengths* (x , y , z) along the directions established by the unit vectors. Lengths are measured from an *origin*, where $(x, y, z) \triangleq (0, 0, 0)$ (Fig. 2.2-1a). The ordering of these axes follows the “right-hand rule”: if the index finger represents $\mathbf{i}$, and the middle finger is bent 90 deg from the index finger to establish the $\mathbf{j}$ direction, then the upward pointing thumb represents the $\mathbf{k}$ direction. Scalar components of the *velocity* can be resolved *along* the $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ directions, while right-handed rotations may occur *around* the $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ axes.

If the origin is unaccelerated (but possibly moving with constant translational velocity) and the unit vectors are not rotating with respect to some absolute reference system of cosmic scale, then this basic frame is an *inertial reference frame*. Most “inertial” frames are not rigorously inertial, but the degree of error in making that assumption is, in some sense, acceptable. For calculating the trajectories of paper clips, or satellites, or interplanetary probes, the most appropriate “inertial” reference plane may be your desk top, or the earth's equator, or the solar system's ecliptic.

Having arrived at a basic reference frame, various *relative frames*, possibly accelerated and rotating, can be defined. These frames allow “local”



Fig. 2.1-3. Boeing 737 flying in the tip-vortex wake of a Lockheed-Martin C-130. (courtesy of NASA)

effects and variations to be described, while the motions of the relative frames (with respect to the basic frame) can be considered separately. Suppose the basic frame is an “inertial” frame whose origin is fixed at a point on the earth’s surface, and an aircraft is flying near the basic frame origin. A relative frame can be fixed at the plane’s center of mass (c.m.). A *body-fixed frame* is one such frame; its origin is at the plane’s c.m., and its axes are aligned with the nose, the right wing tip, and the downward perpendicular to the two. Alternatively, a *velocity-fixed frame* is defined with the c.m. as origin but with one axis aligned to the direction of motion. The second axis may be confined to the horizontal plane, or it may be aligned with the right wing tip. Each of these relative frames has utility in characterizing and analyzing the aircraft’s motions.

TRANSLATIONAL POSITION AND VELOCITY

The *translational (or rectilinear) position* of the center of mass is portrayed by the three components x , y , and z , representing northerly, easterly, and vertical displacement from the origin. By convention, z is positive *down* to preserve a “right-hand-rule” relationship with x and y (Fig. 2.2-1b). The aircraft’s height h above the ground is a positive number; therefore, $h = -z$. The aircraft’s position *vector* $\mathbf{r}$, measured from the origin,

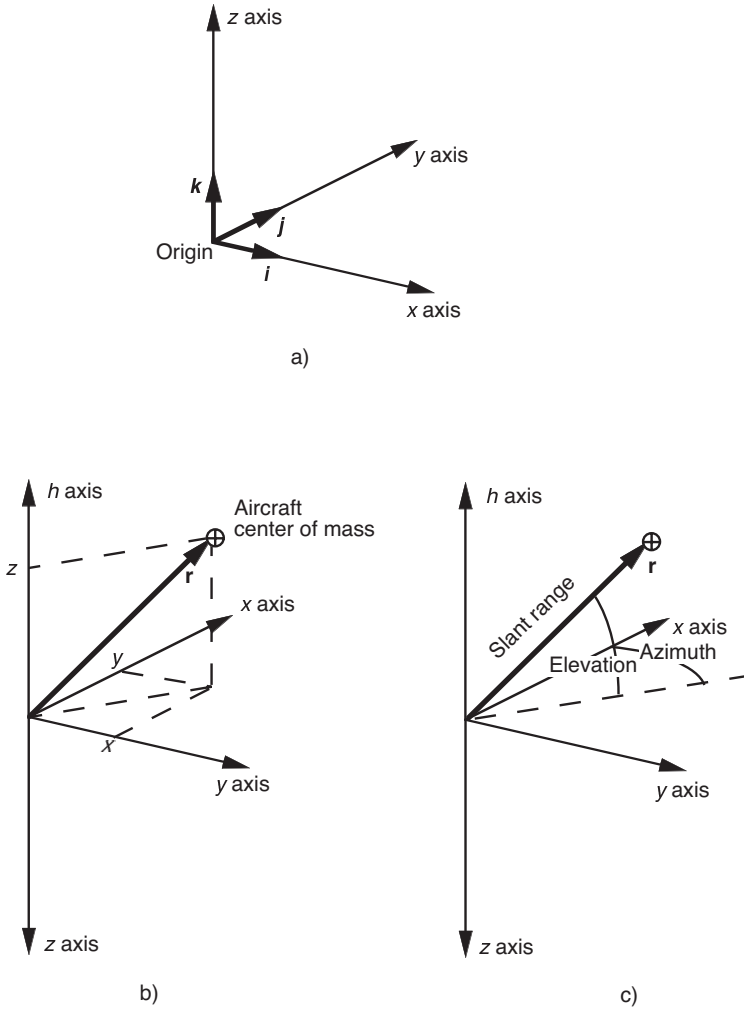


Fig. 2.2-1. Basic Cartesian reference frame. (a) Initial orientation. (b) Conventional orientation used in aircraft dynamics. (c) Corresponding polar coordinates.

is expressed as a column of three elements measured along the i , j , and k directions:

$$\mathbf{r} \triangleq \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} \triangleq \begin{bmatrix} x \\ y \\ z \end{bmatrix} \triangleq \begin{bmatrix} x \\ y \\ -h \end{bmatrix} \quad (2.2-1)$$

x , y , and z , are Cartesian components of $\mathbf{r}$, that is, their axes are orthogonal to each other. The aircraft's *distance from the origin*, r , is the magnitude (or absolute value) of the vector $\mathbf{r}$, calculated as the root sum square of its components:

$$r \triangleq |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2} = \left[\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right]^{1/2} \triangleq [\mathbf{r}^T \mathbf{r}]^{1/2} \quad (2.2-2)$$

The distance is constant if the scalar r is constant. For the position vector to be constant, all three components of $\mathbf{r}$ must not change. This is equivalent to saying that the position vector's magnitude and direction (from the origin) are constant, as $\mathbf{r}$ could be expressed in polar coordinates (Fig. 2.2-1c): *slant range* (= distance), *elevation angle*, and *azimuth angle*.

Equation 2.2-2 introduces the *vector transpose* operation $(\cdot)^T$, which transforms the vector from a column to a row vector, and the *scalar* (or *dot* or *inner*) *product* operation $\mathbf{a}^T \mathbf{b}$, in which the corresponding elements of two equal-dimension vectors are multiplied and summed. Even though $\mathbf{a}$ and $\mathbf{b}$ are vectors, the result is a scalar ($= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$, where n is the number of elements in each vector). The two vectors must be *conformable* for the product to exist, i.e., they must have the same number of elements.

Using MATLAB, eq. 2.2-2 would be expressed as `rmag = sqrt(r' * r)`, where the vector $\mathbf{r}$ would be defined as `[x; y; z]`. MATLAB does not use different notation for vectors and matrices (e.g., boldface or upper case for matrices). The transpose of a vector or matrix is denoted by the prime $(\cdot)'$, whereas this text uses the prime simply to denote a different version of a particular variable. Unless using the Symbolic Toolbox, all of MATLAB's computations are numerical; hence, the x , y , and z that define $\mathbf{r}$ normally would be numerical values.

The aircraft's inertial velocity vector $\mathbf{v}_I$ is the time derivative of $\mathbf{r}$, $d\mathbf{r}/dt$, also symbolized as $\dot{\mathbf{r}}$:

$$\mathbf{v}_I = \dot{\mathbf{r}} \triangleq \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} dx/dt \\ dy/dt \\ dz/dt \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \quad (2.2-3)$$

The time derivative of the vector is a vector of the time derivatives of the scalar Cartesian components. The subscript "I" indicates that the aircraft's velocity (measured relative to the inertial frame) is resolved into

components that are parallel to the bases of the inertial frame of reference (Fig. 2.2-2a). The aircraft's earth-relative speed³ is

$$V = v_I = |\mathbf{v}_I| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \left[\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \right]^{1/2} = [\mathbf{v}_I^T \mathbf{v}_I]^{1/2} \quad (2.2-4)$$

In steady, level flight, h , v_1 , and v_2 are constant, and $v_3 = 0$. If the wind speed is zero, V is also the aircraft's *airspeed*, or speed relative to the surrounding air. If the wind is blowing, the simplest means of expressing the air-relative velocity vector is to add (or subtract, depending on sign conventions) the earth-relative wind velocity components to (from) v_1 , v_2 , and v_3 .

The Cartesian description of the velocity vector can be converted to polar coordinates, with vector magnitude V , *heading angle* (from north) ξ , and *flight path angle* (from the horizontal) γ (Figure 2.2-2a):

$$\mathbf{v}_I \triangleq \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} V \cos \gamma \cos \xi \\ V \cos \gamma \sin \xi \\ -V \sin \gamma \end{bmatrix} \quad (2.2-5)$$

From trigonometry, the inverse relationships for the velocity vector are

$$V = \sqrt{v_1^2 + v_2^2 + v_3^2} \quad (2.2-6)$$

$$\xi = \cos^{-1}[v_1/(v_1^2 + v_2^2)^{1/2}] = \sin^{-1}[v_2/(v_1^2 + v_2^2)^{1/2}] \quad (2.2-7)$$

$$\gamma = \sin^{-1}(-v_3/V) \quad (2.2-8)$$

Consequently, V is always a positive number, ξ does not depend on vertical velocity, and $\gamma = 0$ in steady, level flight. The angles γ and ξ can be depicted as arcs on a sphere surrounding the origin [K-1], as shown in Fig. 2.2-2b.

ANGULAR ORIENTATION AND RATE

The angular orientation of the aircraft's body with respect to the (x' , y' , z') frame (the inertial frame translated to the aircraft's c.m.) is described by three *Euler angles* (Fig. 2.2-3), measured in degrees or radians: ψ (yaw angle), θ (pitch angle), and ϕ (roll angle). Euler angles represent an ordered set of sequential rotations from a reference frame to a frame fixed in the body. The ordering is arbitrary, but once chosen, it must be retained.

3. *Speed* is synonymous with velocity-vector *magnitude*.

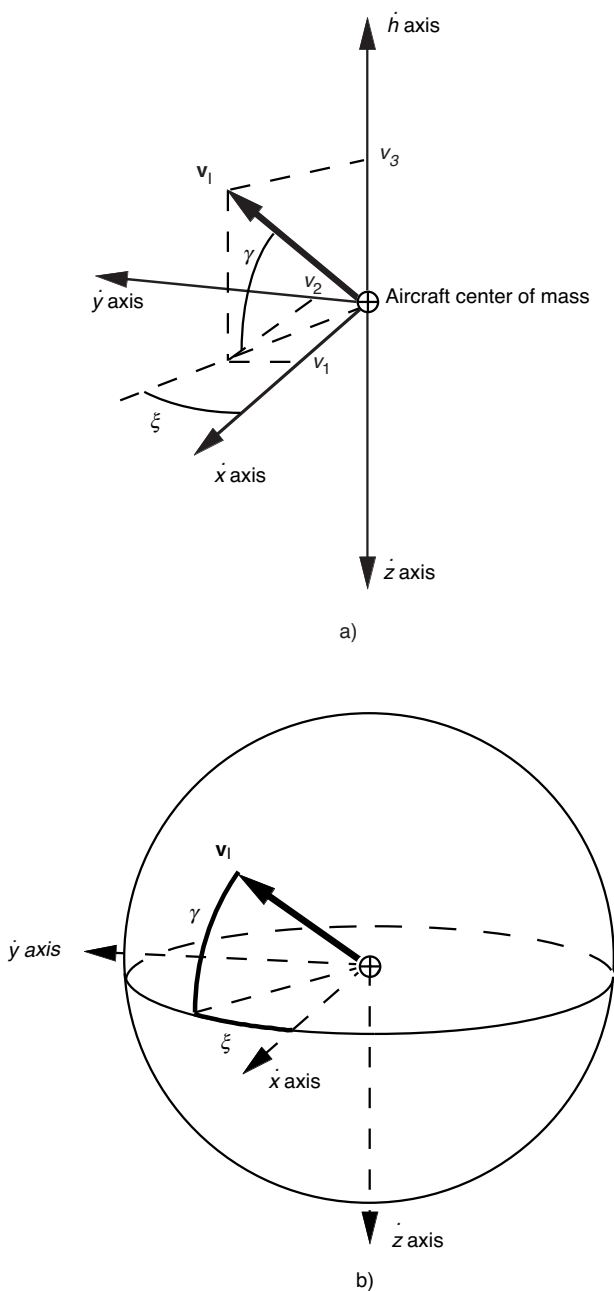


Fig. 2.2-2. Inertial velocity vector $\mathbf{v}_I$ and inertial-frame components (negative value of v_3 shown, corresponding to a positive altitude rate). (a) Definition of polar coordinates of $\mathbf{v}_I$. (b) Projection of γ and ξ on the unit sphere.

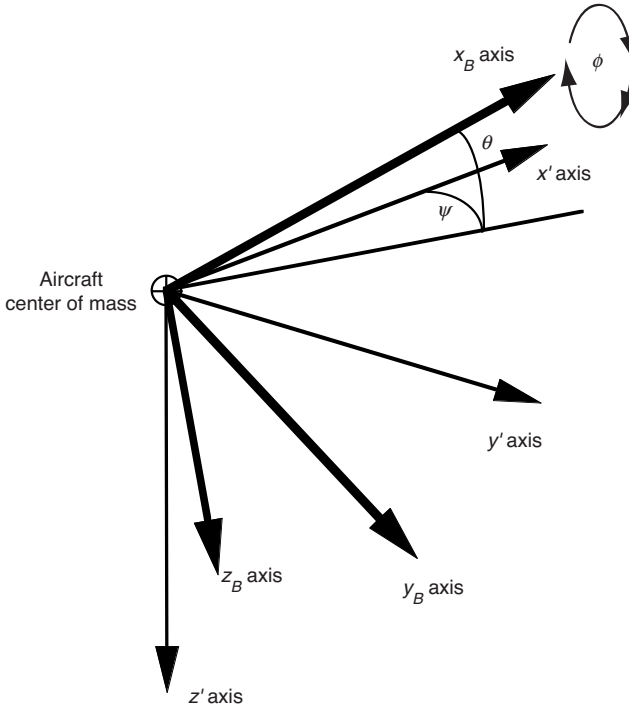


Fig. 2.2-3. Angular orientation of the aircraft relative to the inertial frame. (x_B, y_B, z_B) forms a Cartesian frame referenced to the body, with x_B running along the centerline, y_B pointing to the right, and z_B perpendicular (down) to the first two. The body-frame components of the aircraft's velocity vector (u, v, w) are parallel to (x_B, y_B, z_B) .

Restricting ourselves to right-handed rotations, there are twelve possible rotational sequences. The *conventional sequence* from the inertial frame (Fig. 2.2-4a) to an aircraft's body frame begins with a right-handed *yaw rotation* ψ about the vertical (z') inertial axis, which is also the body (z_B) axis at the reference condition (Fig. 2.2-4b). This is followed by a right-handed *pitch rotation* θ about an inertial axis that is coincident with the intermediate spanwise (y_B) body axis (Fig. 2.2-4c); the y_B axis is still in the horizontal plane, but it is not the same as the y' axis unless $\psi = 0$. A right-handed *roll rotation* ϕ about an inertial axis that is coincident with the body's intermediate centerline (x_B axis) completes the sequence (Fig. 2.2-4d). The basic reference (x', y', z') directions and the body-axis (x_B, y_B, z_B) directions are aligned only when all Euler angles are zero.

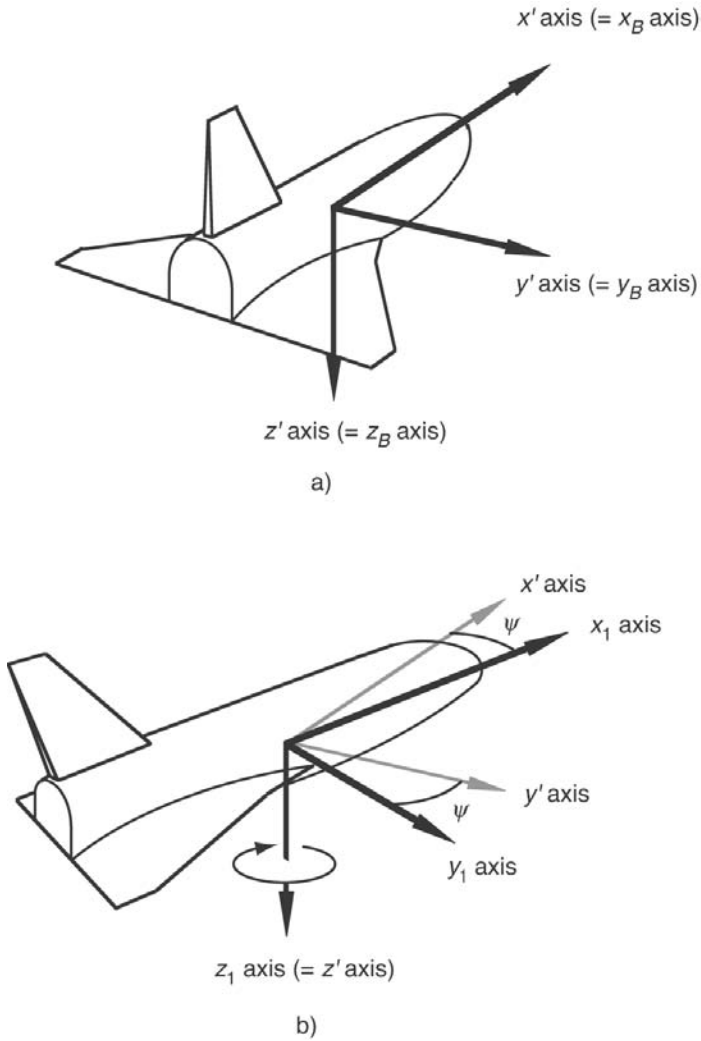


Fig. 2.2-4. Sequence of Euler-angle rotations and their effects on vector transformation. (a) Body frame aligned with inertial frame. (b) Yaw rotation from inertial frame to Intermediate Frame 1.

The three Euler angles are independent of each other, and they can be combined in the *attitude vector* Θ ,

$$\Theta \triangleq \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} \quad (2.2-9)$$

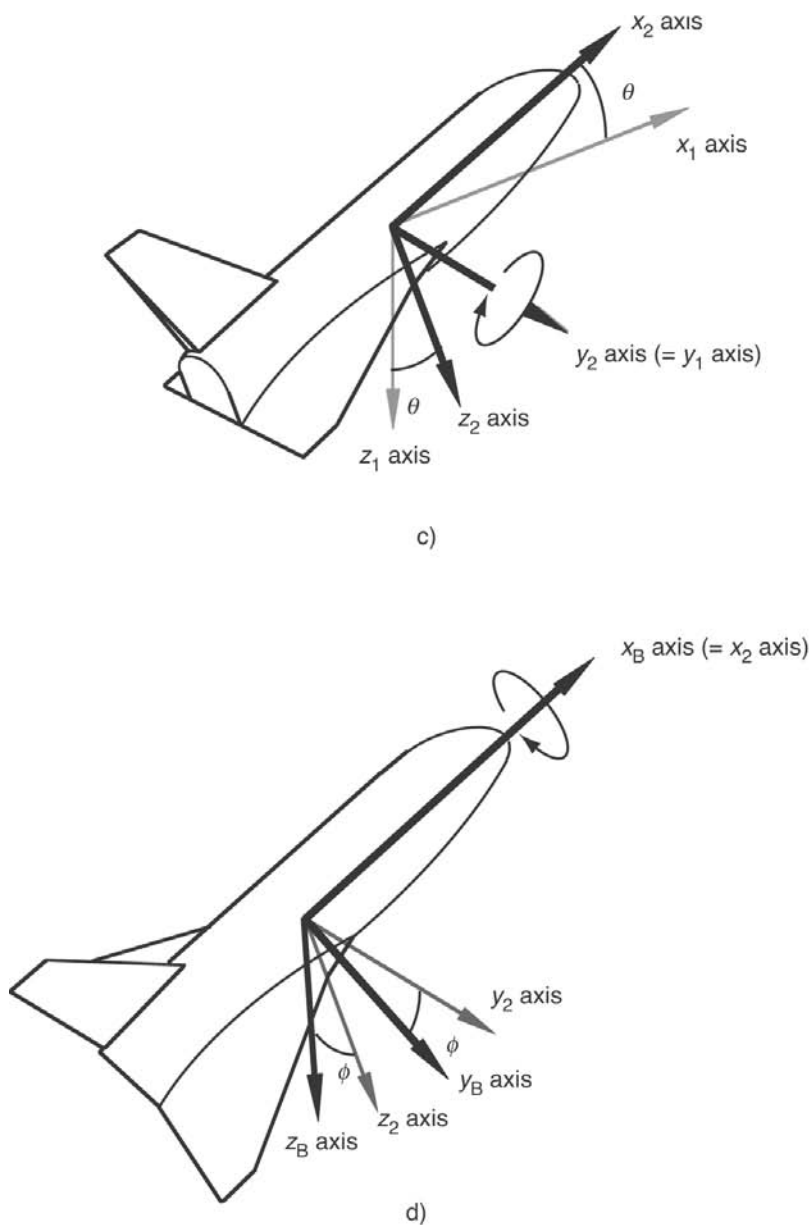


Fig. 2.2-4. *Continued.* Sequence of Euler-angle rotations and their effects on vector transformation. (c) Pitch rotation from Intermediate Frame 1 to Intermediate Frame 2. (d) Roll rotation from Intermediate Frame 2 to body axes.

where the $\Theta = 0$ reference point orients the aircraft with its x axis horizontal and pointing north and with the wings level. A sequence of 45-deg yaw-pitch-roll rotations is mapped on the unit sphere in Fig. 2.2-5; ψ and θ appear as arcs on the surface, while ϕ is represented by a shaded bar twisted away from the horizontal.

Inertial-frame vectors such as the velocity vector often must be resolved into body-fixed coordinates, and the Euler angles are central to making this transformation. $\mathbf{v}_B$ is the inertial velocity of the aircraft expressed in body-axis components:

$$\mathbf{v}_B \triangleq \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \mathbf{H}_I^B \mathbf{v}_I \quad (2.2-10)$$

Here $\mathbf{v}_I$ is given by eq. 2.2-3 or 2.2-5, and $\mathbf{H}_I^B$ is a (3×3) matrix. (Note that $\mathbf{v}_B$ and $\mathbf{v}_I$ are the same vector observed in different axis systems.) More particularly, $\mathbf{H}_I^B$ is an *orthonormal transformation* from the inertial frame to the body frame. It transforms the components of a vector expressed in one orthogonal frame to the vector's components in another orthogonal frame while preserving the vector's magnitude:

$$\mathbf{H}_I^B = \begin{bmatrix} \cos\theta \cos\psi & \cos\theta \sin\psi & -\sin\theta \\ (-\cos\phi \sin\psi + \sin\phi \sin\theta \cos\psi) & (\cos\phi \cos\psi + \sin\phi \sin\theta \sin\psi) & \sin\phi \cos\theta \\ (\sin\phi \sin\psi + \cos\phi \sin\theta \cos\psi) & (-\sin\phi \cos\psi + \cos\phi \sin\theta \sin\psi) & \cos\phi \cos\theta \end{bmatrix} \quad (2.2-11)$$

The *matrix-vector product* in eq. 2.2-10 is a vector of the dot products of the matrix's rows with the column vector. Given an $(m \times n)$ matrix and an $(n \times 1)$ vector, the product is an $(m \times 1)$ vector. In this case, $m = n = 3$; therefore, u is given by

$$u = v_1 \cos\theta \cos\psi + v_2 \cos\theta \sin\psi - v_3 \sin\theta \quad (2.2-12)$$

and v and w are calculated accordingly.

The elements of $\mathbf{H}_I^B$ depend on the specified sequence of Euler angle rotations: yaw, pitch, roll. Yaw rotation from the Inertial Frame through the angle ψ to *Intermediate Frame 1* (Fig. 2.2-4b) produces the transformation matrix $\mathbf{H}_I^1$:

$$\mathbf{H}_I^1 \triangleq \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2.2-13)$$

The first row of $\mathbf{H}_I^1$ projects the three elements of an inertial-frame vector onto the first element of a Frame 1 vector, and so on. The velocity vector resolved into Frame 1 coordinates is related to the inertial-frame expression as follows:

$$\mathbf{v}_1 \triangleq \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_1 = \mathbf{H}_I^1 \mathbf{v}_I = \mathbf{H}_I^1 \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_I = \begin{bmatrix} v_1 \cos\psi + v_2 \sin\psi \\ -v_1 \sin\psi + v_2 \cos\psi \\ v_3 \end{bmatrix} \quad (2.2-14)$$

The pitch rotation from Intermediate Frame 1 through the angle θ to *Intermediate Frame 2* (Fig. 2.2-4c) produces the transformation matrix $\mathbf{H}_1^2$:

$$\mathbf{H}_1^2 \triangleq \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \quad (2.2-15)$$

which operates on $\mathbf{v}_1$ to produce $\mathbf{v}_2$:

$$\begin{aligned} \mathbf{v}_2 &= \mathbf{H}_1^2 \mathbf{v}_1 = \mathbf{H}_1^2 \mathbf{H}_I^1 \mathbf{v}_I \\ &= \begin{bmatrix} (v_1 \cos\psi + v_2 \sin\psi) \cos\theta - v_3 \sin\theta \\ (-v_1 \sin\psi + v_2 \cos\psi) \\ -(v_1 \cos\psi + v_2 \sin\psi) \sin\theta + v_3 \cos\theta \end{bmatrix} \end{aligned} \quad (2.2-16)$$

Alternatively, the same result is obtained through multiplying $\mathbf{v}_I$ by $\mathbf{H}_I^2$, defined as

$$\mathbf{H}_I^2 \triangleq \mathbf{H}_1^2 \mathbf{H}_I^1 = \begin{bmatrix} \cos\theta \cos\psi & \cos\theta \sin\psi & -\sin\theta \\ -\sin\psi & \cos\psi & 0 \\ -\sin\theta \cos\psi & \sin\theta \sin\psi & \cos\theta \end{bmatrix} \quad (2.2-17)$$

where the rows of $\mathbf{H}_1^2$ multiply the columns of $\mathbf{H}_I^1$ to form individual elements of the *matrix product* $\mathbf{H}_I^2$.

The rotational sequence is completed by the roll rotation from Frame 2 through the angle ϕ to the body frame (Fig. 2.2-4d). The transformation matrix $\mathbf{H}_2^B$ does the job, with

$$\mathbf{H}_2^B \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \quad (2.2-18)$$

$\mathbf{H}_1^B$ is then defined by the triple-matrix product

$$\mathbf{H}_1^B(\phi, \theta, \psi) = \mathbf{H}_2^B(\phi)\mathbf{H}_1^2(\theta)\mathbf{H}_1^1(\psi) \quad (2.2-19)$$

to yield eq. 2.2-11. A corresponding MATLAB equation involving matrix products is $\mathbf{H1B} = \mathbf{H2B} * \mathbf{H12} * \mathbf{H11}$.

If $\mathbf{v}_B$ were known and it was necessary to know $\mathbf{v}_I$, the inverse to eq. 2.2-10 could be invoked:

$$\mathbf{v}_I = \mathbf{H}_B^I \mathbf{v}_B \quad (2.2-20)$$

where $\mathbf{H}_B^I$ is the *matrix inverse* of $\mathbf{H}_1^B$, that is, $\mathbf{H}_B^I \triangleq (\mathbf{H}_1^B)^{-1}$. Because $\mathbf{H}_1^B$ is a square, orthonormal matrix, its inverse is its *matrix transpose* $(\mathbf{H}_1^B)^T$, formed by interchanging the rows and columns of the original matrix:

$$\mathbf{H}^T = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}^T \triangleq \begin{bmatrix} h_{11} & h_{21} & h_{31} \\ h_{12} & h_{22} & h_{32} \\ h_{13} & h_{23} & h_{33} \end{bmatrix} \quad (2.2-21)$$

The *Euler-angle rate vector* $\dot{\Theta}$ is simply the term-by-term time derivative of the Euler-angle vector,

$$\dot{\Theta} \triangleq \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \quad (2.2-22)$$

Like the Euler-angle vector, the components of the Euler-angle rate vector are not measured along orthogonal axes. Unlike the translational velocity vector, transforming it into different axis systems is a complicated process. Nevertheless, it is possible to express the angular rate as an orthogonal vector; for example, the inertial angular rates measured about the body's (x_B, y_B, z_B) axes are defined by

$$\omega_B = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (2.2-23)$$

The inertial angular rate vector observed in the body-axis frame is related to the Euler angle rate vector by the nonorthogonal transformation $\omega_B = \mathbf{L}_E^B \dot{\Theta}$, where

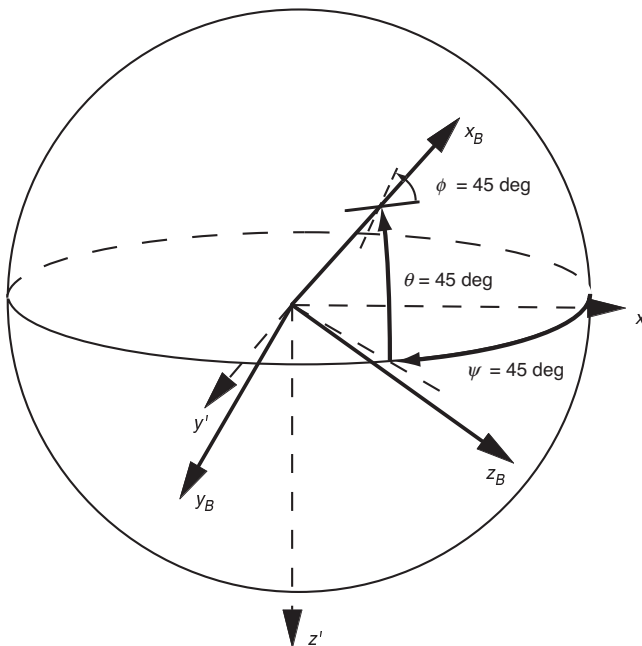


Fig. 2.2-5. Projection of the aircraft's Euler angles on the unit sphere.

$$\mathbf{L}_E^B \triangleq \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \quad (2.2-24)$$

$\mathbf{L}_E^B$ represents separate transformations of the three Euler angle rate components by three different orthonormal matrices, the first of which is an identity matrix⁴:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \mathbf{I}_n \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + \mathbf{H}_2^B \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \mathbf{H}_2^B \mathbf{H}_1^2 \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} \quad (2.2-25)$$

4. The *identity matrix* $\mathbf{I}_n$ is an $(n \times n)$ matrix containing 1's on its principal diagonal (from upper left to lower right corners) and zeros elsewhere. Multiplying a vector by $\mathbf{I}_n$ does not change the vector in any way; hence, it provides an orthonormal transformation by default.

Only the first column of $\mathbf{I}_n$, the second column of $\mathbf{H}_2^B$, and the third column of $\mathbf{H}_2^B \mathbf{H}_1^2$ are used to define $\mathbf{L}_E^B$, as the remaining columns multiply zero vector elements in eq. 2.2-25.

The individual transformations are orthonormal, but the complete transformation is not; hence, the inverse of $\mathbf{L}_E^B$ is not simply its transpose. It is

$$\mathbf{L}_B^E \triangleq (\mathbf{L}_E^B)^{-1} = \begin{bmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{bmatrix} \quad (2.2-26)$$

where $\mathbf{A}^{-1} = \text{Adj}(\mathbf{A})/\text{Det}(\mathbf{A})$, as detailed below. The orthonormal transformations presented earlier do not have singularities, but $\mathbf{L}_B^E$ has four infinite elements when $\theta = \pm 90$ deg, that is, when the airplane's nose is pointed straight up or down.

The *inverse* of an $(n \times n)$ matrix is the $(n \times n)$ adjoint of the matrix divided by the scalar matrix determinant. The *determinant* is the sum of various permutations of products of the matrix elements. For $n = 3$,

$$\text{Det}(\mathbf{A}) = |\mathbf{A}| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \quad (2.2-27)$$

For $n > 3$, Laplace expansion or pivotal condensation can be used to find the determinant (e.g., [S-17]). The *adjoint matrix* is the transpose of the matrix of cofactors of the original matrix,

$$\text{Adj}(\mathbf{A}) = \mathbf{C}^T \quad (2.2-28)$$

where the *cofactors* are signed minors of the original matrix. The *minor* of element a_{ij} is the determinant of the matrix formed by the elimination of the i th row and j th column of $\mathbf{A}$; the cofactor's sign is positive if $(i + j)$ is even, negative if $(i + j)$ is odd. For $n = 3$,

$$\text{Adj}(\mathbf{A}) = \begin{bmatrix} (a_{22}a_{33} - a_{23}a_{32}) & (a_{13}a_{32} - a_{12}a_{33}) & (a_{12}a_{23} - a_{13}a_{22}) \\ (a_{23}a_{31} - a_{21}a_{33}) & (a_{11}a_{33} - a_{13}a_{31}) & (a_{13}a_{21} - a_{11}a_{23}) \\ (a_{21}a_{32} - a_{22}a_{31}) & (a_{12}a_{31} - a_{11}a_{32}) & (a_{11}a_{22} - a_{12}a_{21}) \end{bmatrix} \quad (2.2-29)$$

This symbolic representation of the inverse is useful in later chapters; however, if only numerical values are required, MATLAB calculates the inverse and determinant as `inv(A)` and `det(A)`.

$\boldsymbol{\omega}_B$ can be transformed to an orthogonal angular rate vector in the inertial frame, $\boldsymbol{\omega}_I = \mathbf{H}_B^I \boldsymbol{\omega}_B$, where the elements of $\boldsymbol{\omega}_I$ are measured about the orthogonal inertial axes; therefore $\boldsymbol{\omega}_I$ does not equal $\dot{\boldsymbol{\Theta}}$. The inertial angular rate vector $\boldsymbol{\omega}_I$ has use in the description of steady, coordinated turns (Section 2.6).

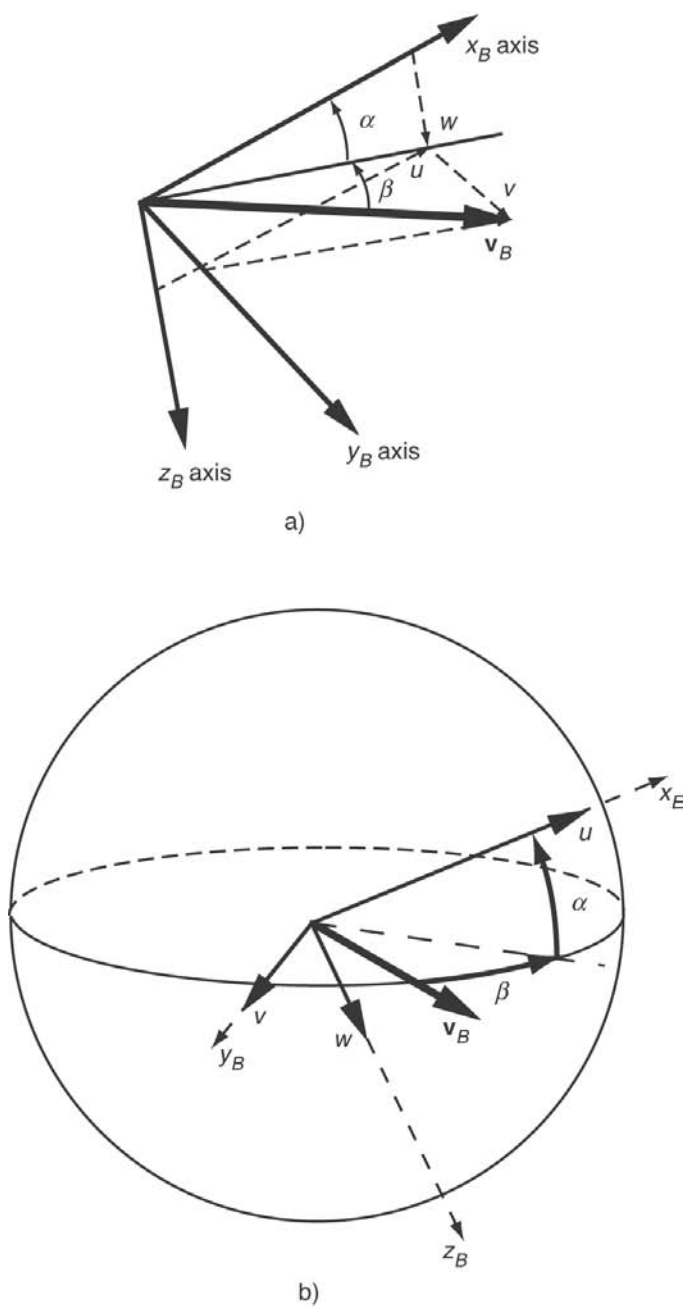


Fig. 2.2-6. Relationships between inertial velocity ($\mathbf{v}_B$), body-frame components (u , v , w), and airflow angles (α , β). The magnitude of $\mathbf{v}_B$ is V , assuming that the wind speed is zero. (a) Definition of polar coordinates of $\mathbf{v}_B$. (b) Projection of α and β on the unit sphere.

AIRFLOW ANGLES

The orientation of the velocity vector with respect to body axes plays an important role in determining the aircraft's aerodynamic forces and moments. Two angles express this orientation: the *sideslip angle* β and the *angle of attack* α (Fig. 2.2-6). They are defined such that

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} V \cos \alpha \cos \beta \\ V \sin \beta \\ V \sin \alpha \cos \beta \end{bmatrix} \quad (2.2-30)$$

Therefore, a little trigonometry yields the inverse relationships:

$$\begin{bmatrix} V \\ \beta \\ \alpha \end{bmatrix} = \begin{bmatrix} \sqrt{u^2 + v^2 + w^2} \\ \sin^{-1}(v/V) \\ \tan^{-1}(w/u) \end{bmatrix} \quad (2.2-31)$$

Equations 2.2-6 and 2.2-31 give alternative definitions for airspeed V when the wind speed is zero. When the wind speed is not zero, it is important to distinguish between air-relative and earth-relative velocity vectors, as discussed in the next chapter.

Vectors expressed in wind axes can be represented in body axes, and vice versa; hence, another transformation matrix is needed. Initially aligning the aircraft with the velocity vector, the wind-relative attitude is defined by a *left-handed* rotation through the sideslip angle β , then a right-handed rotation through the angle of attack α . The transformation matrix $\mathbf{H}_W^B$ is thus defined as

$$\begin{aligned} \mathbf{H}_W^B &\triangleq \mathbf{H}_3^B(\alpha) \mathbf{H}_W^3(-\beta) = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos \alpha \cos \beta & -\cos \alpha \sin \beta & -\sin \alpha \\ \sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & -\sin \alpha \sin \beta & \cos \alpha \end{bmatrix} \quad (2.2-32) \end{aligned}$$

SUMMARY OF AXIS SYSTEMS AND TRANSFORMATIONS

Four axis systems have value in the dynamic analysis of aircraft:

- Inertial axes
- Body axes

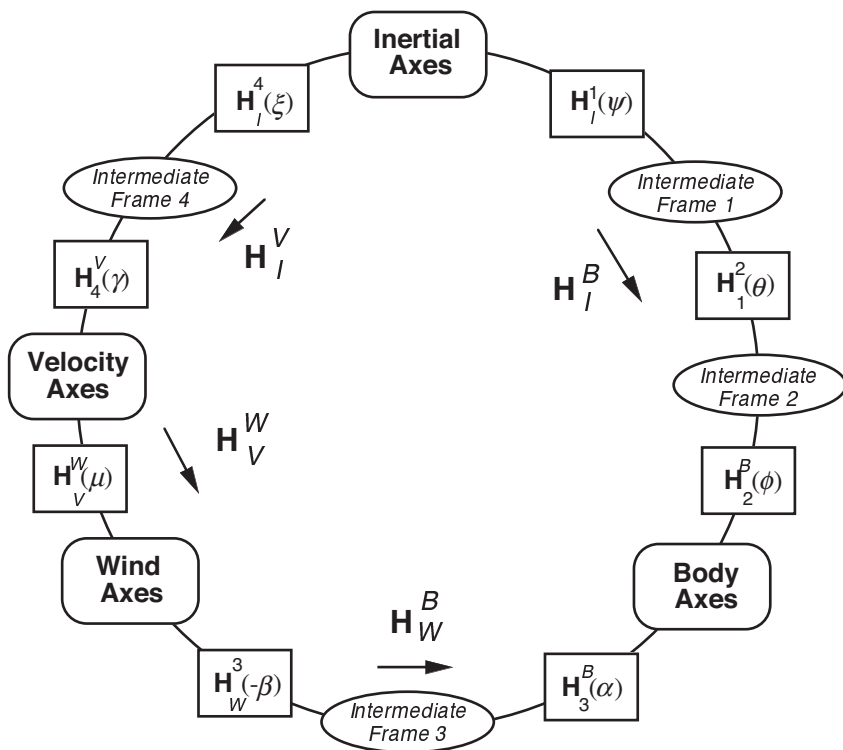


Fig. 2.2-7. Reference-axis transformations. Arrows indicate right-hand rotations. (adapted from [S-15]).

- Wind axes
- Velocity axes

Figure 2.2-7, adapted from [S-15], illustrates that each axis system is related to the others by a series of orthonormal transformations. Inertial-frame vectors are transformed to the body frame by H_I^B (eq. 2.2-11). Vectors expressed in the body frame are transformed to the wind-axis frame by H_B^W , the inverse (and transpose) of eq. 2.2-32. Thus, a transformation from inertial to wind axes can be established through the body frame of reference as $H_I^W \triangleq H_B^W(-\beta, \alpha)H_I^B(\phi, \theta, \psi)$.

Alternatively, H_I^W could be written as a function of the vertical and horizontal flight path angles γ and ξ as well as the *bank angle* μ . In typical

flight conditions, the bank angle is very nearly the same as the roll angle; however, the former is measured about the velocity vector while the latter is measured about the body x axis. Using eq. 2.2-5 and Fig. 2.2-2, the transformation from inertial to velocity axes is

$$\mathbf{H}_I^V \triangleq \mathbf{H}_4^V(\gamma)\mathbf{H}_I^4(\xi) = \begin{bmatrix} \cos \gamma & 0 & -\sin \gamma \\ 0 & 1 & 0 \\ \sin \gamma & 0 & \cos \gamma \end{bmatrix} \begin{bmatrix} \cos \xi & \sin \xi & 0 \\ -\sin \xi & \cos \xi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2.2-33)$$

Transformation through the bank angle rotation completes the alternate route,

$$\mathbf{H}_I^W(\mu, \gamma, \xi) \triangleq \mathbf{H}_V^W(\mu)\mathbf{H}_I^V(\gamma, \xi) \quad (2.2-34)$$

where

$$\mathbf{H}_V^W \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \mu & \sin \mu \\ 0 & -\sin \mu & \cos \mu \end{bmatrix} \quad (2.2-35)$$

The various rotations are brought together by their projections on the unit sphere (Fig. 2.2-8), which relate the inertial x' direction to the vehicle's centerline direction and to the direction of travel. Note in particular the distinction between the roll angle ϕ (about the centerline) and the bank angle μ (about the velocity vector). Using spherical trigonometry, the bank angle can be derived from the remaining flight path and aerodynamic angles [K-1]:

$$\mu = \sin^{-1} \left\{ \frac{[\cos \theta \sin \phi \cos \beta + (\cos \alpha \sin \theta - \sin \alpha \cos \theta \cos \phi) \sin \beta]}{\cos \gamma} \right\} \quad (2.2-36a)$$

or

$$\mu = \cos^{-1} \left[\frac{\cos \alpha \cos \theta \cos \phi + \sin \alpha \sin \theta}{\cos \gamma} \right] \quad (2.2-36b)$$

These equations can also be derived by equating the (3, 2) and (3, 3) elements of

$$\mathbf{H}_W^V = \mathbf{H}_I^V \mathbf{H}_B^I \mathbf{H}_W^B \quad (2.2-37)$$

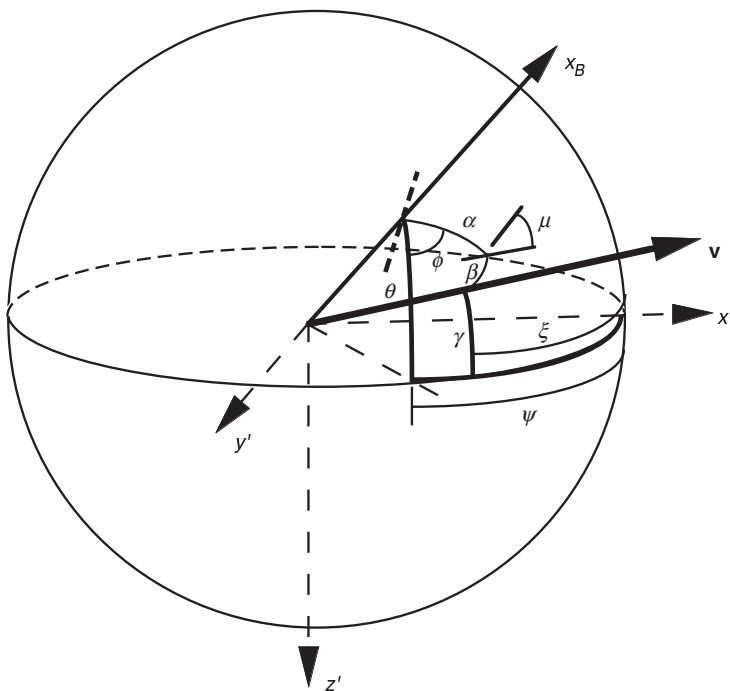


Fig. 2.2-8. Projections of flight path, airflow, and Euler angles on the unit sphere.

With the wing tips level, there are two simple relationships among the flight-path, aerodynamic, and body-attitude angles (Fig. 2.2-9). In the vertical plane, the flight-path angle is the difference between the pitch angle and the angle of attack:

$$\gamma = \theta - \alpha \quad (2.2-38)$$

In the horizontal plane and with horizontal velocity, the flight-path (headings) angle is the sum of the yaw angle and the sideslip angle:

$$\xi = \psi + \beta \quad (2.2-39)$$

Example 2.2-1. Flight Path and Aerodynamic Angles of a Jet Transport

Consider a jet transport that is maneuvering from level flight into a descending banked turn. It is flying in still air, with instantaneous inertial

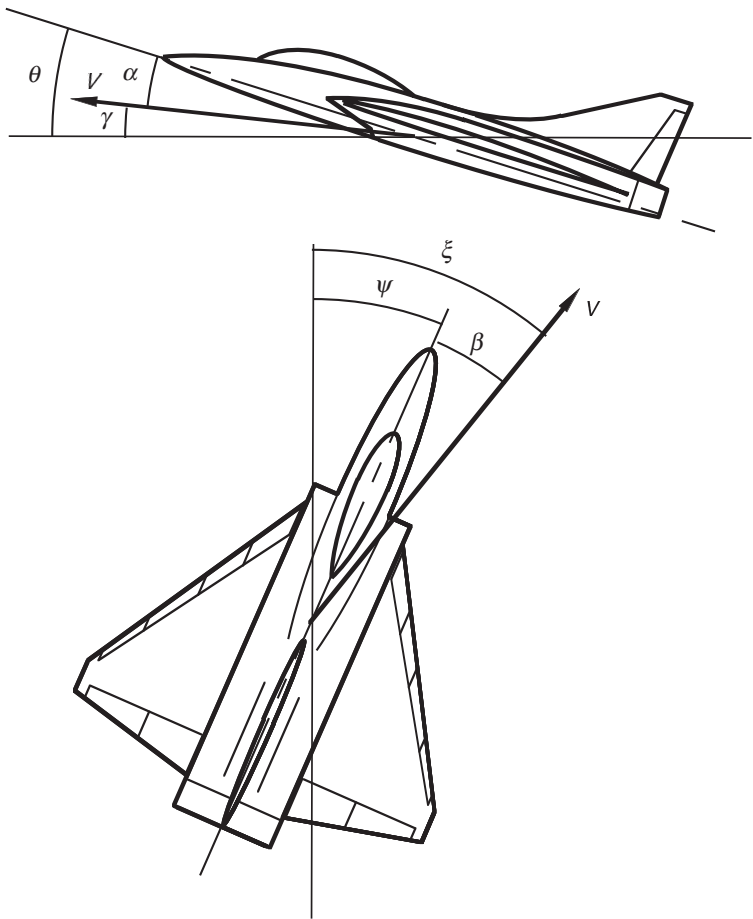


Fig. 2.2-9. Relationship among flight path, aerodynamic, and body-attitude angles with level wing tips.

velocity components and Euler angles

$$\mathbf{v}_I = \begin{bmatrix} 200 \\ 10 \\ 5 \end{bmatrix} \text{ m/s}; \quad \boldsymbol{\Theta} = \begin{bmatrix} 10 \\ 10 \\ 2 \end{bmatrix} \text{ deg}$$

The flight path and aerodynamic angles at this point on the aircraft's trajectory are calculated by the following MATLAB code:

```

vi   = [200; 10; 5];           % Inertial velocity in inertial frame
dr   = pi / 180;               % Degrees-to-radian conversion
th   = [10 * dr; 10 * dr; 2 * dr] % Euler angles entered as a column vector
V    = sqrt(vi' * vi)          % Airspeed (m/s)
gam  = asin(-vi(3) / V);        % Vertical flight path angle (rad)
gamd = gam / dr                 % Vertical flight path angle (deg)
xi   = asin(vi(2) / sqrt(vi(1)^2 + vi(2)^2)); % Horizontal flight path angle (rad).
xid  = xi / dr                  % Horizontal flight path angle (deg)
HI1  = [cos(th(3)) sin(th(3)) 0; -sin(th(3)) cos(th(3)) 0; 0 0 1] % Yaw transformation matrix
H12  = [cos(th(2)) 0 -sin(th(2)); 0 1 0; sin(th(2)) 0 cos(th(2))]; % Pitch transformation matrix
H2B  = [1 0 0; 0 cos(th(1)) sin(th(1)); 0 -sin(th(1)) cos(th(1))]; % Roll transformation matrix
HIB  = H2B * H12 * HI1          % Inertial-to-body-axis transformation
vb   = HIB * vi                 % Inertial velocity, body-axis frame
be   = asin(vb(2) / V);         % Sideslip angle (rad)
bed  = be / dr                  % Sideslip angle (deg)
al   = atan(vb(3) / vb(1));      % Angle of attack (rad)
ald  = al / dr                  % Angle of attack (deg)
mu   = asin(cos(th(2)) * sin(th(1)) * cos(be) + (cos(al) * sin(th(2)) ..
    - sin(al) * cos(th(2)) * cos(th(1))) * sin(be)/cos(gam)); % Bank angle (rad)
mud  = mu / dr                  % Bank angle (deg)

```

The MATLAB m-file output is

```

th   =
    0.1745
    0.1745
    0.0349
V    =
   200.3123
gamd =
   -1.4303
xid  =
    2.864
HI1  =
    0.9994    0.0349    0
   -0.0349    0.9994    0
    0         0         1
HIB  =
    0.9842    0.0344   -0.1736
   -0.0042    0.9853    0.1710
    0.1770   -0.1676    0.9698
vb   =
   196.3170
    9.8609
   38.5667
bed  =
    2.8217
ald  =
   11.1143
mud  =
    9.7902

```

Example 2.2-2. Angular Rates of the Jet Transport

Suppose that the aircraft's body-axis angular rates in the previous example are

$$\omega_B = \begin{bmatrix} 10 \\ 4 \\ 2 \end{bmatrix} \text{ deg/s}$$

What are the corresponding Euler-angle rates?

Equation 2.2-26 could be used to write L_E^B directly; then $\dot{\Theta} = L_E^B \omega_B$. However, for illustration, L_B^E is calculated using eq. 2.2-24, and the inverse is evaluated numerically by MATLAB:

m file

```
dr = pi / 180;
th = [10 * dr; 10 * dr; 2 * dr];
om = dr * [10; 4; 2]; % Body-axis rates converted to radians
LEB = [1 0 -sin(th(2)); 0 cos(th(1)) sin(th(1)) * cos(th(2)); ...
       0 -sin(th(1)) cos(th(1)) * cos(th(2))];
      % Euler-rate-to-body-rate transformation
LBE = inv(LEB) % Inverse of LEB matrix
eul = LBE * om / dr % Euler-angle rates (deg)
```

Output

```
LEB =
    1.0000    0    -0.1736
    0    0.9848    0.1710
    0   -0.1736    0.9698

LBE =
    1.0000    0.0306    0.1736
    0    0.9848   -0.1736
    0    0.1763    1.0000

eul =
    10.4698
     3.5919
     2.7053
```

2.3 Forces and Moments

A *force* is a physical influence that changes the translational motion of an unrestrained object, such as a flying aircraft. A *moment* (or *torque*) is the

result of a force being applied at right angles to a *moment arm* that defines the distance from a rotational center to the force. The forces and moments acting on the aircraft are three-dimensional orthogonal vectors $\mathbf{f}$ and $\mathbf{m}$ that can be expressed in alternate frames according to their use. Thus, a force expressed in an inertial frame is

$$\mathbf{f}_I \triangleq \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_I \quad (2.3-1)$$

while the moment of that force expressed in the same frame is

$$\mathbf{m}_I \triangleq \mathbf{r}_I \times \mathbf{f}_I \triangleq \tilde{\mathbf{r}}_I \mathbf{f}_I = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}_I \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_I = \begin{bmatrix} -zY + yZ \\ zX - xZ \\ -yX + xY \end{bmatrix}_I \quad (2.3-2)$$

$\mathbf{r}_I \triangleq [x \ y \ z]_I^T$ is the moment-arm vector between the rotational center and the point of force application. $\mathbf{r}_I \times \mathbf{f}_I$ is the *cross product* of the two vectors $\mathbf{r}_I$ and $\mathbf{f}_I$, and $\tilde{\mathbf{r}}_I$ is the *cross-product-equivalent matrix* of the moment-arm vector. The units of force are newtons,⁵ so newton-meters are appropriate units for moments. The MATLAB function `cross` produces the numerical result: `mI = cross (rI, fI)`.

ALTERNATIVE AXIS SYSTEMS

Forces and moments expressed in noninertial frames can be referred to an inertial frame using the orthonormal transformations of Section 2.2. *Drag* is a force component that is parallel and opposed to the velocity vector, while *lift* is a force component perpendicular to the velocity vector and to the y_B axis; thus, these force components are wind-axis quantities. Yawing, pitching, and rolling moments are most directly associated with the body axes.

Consider the mapping of forces from wind to inertial axes in a two-step process. The drag (D), side force (SF), and lift (L) transform into body-axis axial, lateral, and vertical aerodynamic force components (X , Y , Z):

$$\mathbf{f}(\text{aero})_B = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_B = \mathbf{H}_W^B \begin{bmatrix} -D \\ SF \\ -L \end{bmatrix} = \mathbf{H}_W^B \mathbf{f}(\text{aero})_W \quad (2.3-3)$$

5. One *newton* is the force required to accelerate one kilogram of mass at one meter per second per second.

The minus signs account for normal conventions (drag is positive aft, side force is positive to the right, and lift is positive up). Inertial-frame forces can then be written as

$$\mathbf{f}(\text{aero})_I = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_I = \mathbf{H}_B^I \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_B = \mathbf{H}_B^I \mathbf{f}(\text{aero})_B = \mathbf{H}_W^I \begin{bmatrix} -D \\ SF \\ -L \end{bmatrix} \quad (2.3-4)$$

Body-axis roll, pitch, and yaw moments (L , M , N) can be transformed to the inertial frame, as needed for the examination of steady turns in Section 2.3:

$$\mathbf{m}(\text{aero})_I = \begin{bmatrix} L \\ M \\ N \end{bmatrix}_I = \mathbf{H}_B^I \begin{bmatrix} L \\ M \\ N \end{bmatrix}_B = \mathbf{H}_B^I \mathbf{m}(\text{aero})_B \quad (2.3-5)$$

The accepted symbols for lift force and roll moment are, unfortunately, the same; however, they can usually be differentiated by usage.

AERODYNAMIC FORCES AND MOMENTS

The scalar force and moment components can be written as products of nondimensional coefficients and dimensionalizing quantities. Nondimensional coefficients facilitate the comparison of aerodynamic data for various aircraft configurations, and they allow subscale model wind tunnel data to be applied to full-scale aircraft. The dimensionalizing quantities for forces are the *dynamic pressure* $\bar{q}$ (n/m²) and a *reference area* S (m²). Thus, the drag, side force, and lift can be expressed as

$$D = C_D \bar{q} S \quad (2.3-6)$$

$$SF = C_{SF} \bar{q} S \quad (2.3-7)$$

$$L = C_L \bar{q} S \quad (2.3-8)$$

The reference area is normally the aircraft's wing area, while the dynamic pressure is

$$\bar{q} \triangleq \frac{\rho V_A^2}{2} \quad (2.3-9)$$

where ρ , the air density (kg/m³), is a function of altitude. V_A is the *true airspeed* (or TAS), which equals the earth-relative speed V when the wind is zero.

Three additional definitions of airspeed can be found [K-2]. The *indicated airspeed* (IAS) is derived from the difference in stagnation and static pressure measurements made by a pitot tube and a static pressure port. (Static pressure also defines *pressure altitude*.) The *calibrated airspeed* (CAS) is the IAS corrected for instrumentation and position errors. The *equivalent airspeed* (EAS) is the CAS corrected for compressibility effects on the measurements. For a given dynamic pressure, TAS changes with altitude, while EAS is constant. Thus, EAS and TAS are the same at sea level, and $EAS = TAS \sqrt{\rho(z)\rho_{SL}}$ at altitude z . In common usage, the EAS and CAS are often called IAS.

Aerodynamic force coefficients are sensitive to the airplane's airflow angles. At low airflow angles and zero angular rates, the total lift coefficient due to the wing, fuselage, and horizontal tail is approximately linear in angle of attack:

$$C_L = C_{L_o} + C_{L_\alpha} \alpha \quad (2.3-10)$$

where the coefficients depend on the aircraft's configuration—notably the wing planform, chord section, and twist [A-6]—as well as the flight condition and control settings (Section 2.4). C_{L_o} is the lift coefficient at zero angle of attack, and the *lift slope* C_{L_α} is the partial derivative of the lift coefficient, evaluated at zero angle of attack:

$$C_{L_\alpha} \triangleq \left. \frac{\partial C_L}{\partial \alpha} \right|_{\alpha=0^\circ} \quad (2.3-11)$$

At high α , the wing stalls, limiting the lift coefficient to a maximum value, and this linear lift characteristic breaks down (Fig. 2.3-1). The *angle of attack for zero lift*, α_{zl} , is

$$\alpha_{zl} = -C_{L_o}/C_{L_\alpha} \quad (2.3-12)$$

C_{L_o} is usually positive, and α_{zl} is typically less than zero.

Similarly, the side-force coefficient can be written as a function of sideslip angle,

$$C_{SF} = C_{SF_o} + C_{SF_\beta} \beta \quad (2.3-13)$$

For most aircraft and intended flight conditions, side-force coefficients and derivatives are small.

The drag coefficient is very nearly proportional to the square of the lift coefficient at low airflow angles (Fig. 2.3-1):

$$C_D = C_{D_o} + \epsilon C_L^2 \quad (2.3-14a)$$

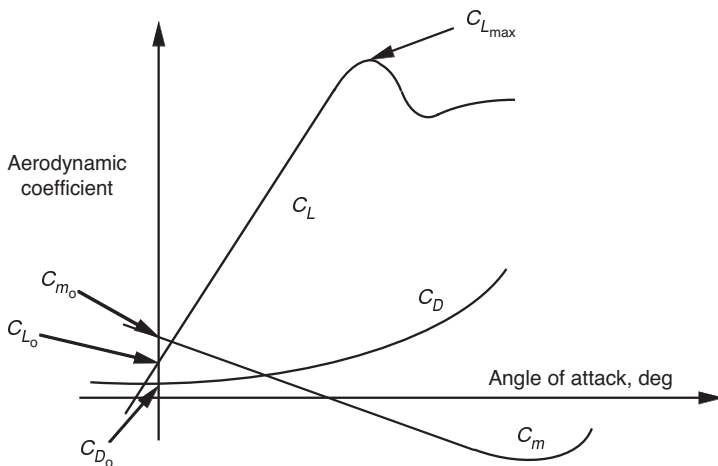


Fig. 2.3-1. Typical variations of lift coefficient C_L , drag coefficient C_D , pitching moment coefficient C_m , and lift-drag ratio C_L/C_D with angle of attack α .

where ε is an *induced drag factor* resulting from the efficiency with which lift is created [A-3]. From eq. 2.3-10, this leads to

$$C_D = (C_{D_0} + \varepsilon C_{L_0}^2) + C_{D_\alpha} + C_{D_{\alpha^2}} \alpha^2 \quad (2.3-14b)$$

where

$$C_{D_\alpha} \triangleq \left. \frac{\partial C_D}{\partial \alpha} \right|_{\alpha=0^\circ} = 2\varepsilon C_{L_0} C_{L_\alpha} \quad (2.3-15)$$

$$C_{D_{\alpha^2}} \triangleq \left. \frac{\partial C_D}{\partial (\alpha^2)} \right|_{\alpha=0^\circ} = \varepsilon C_{L_{\alpha^2}} \quad (2.3-16)$$

The *lift-drag ratio* C_L/C_D or L/D is a measure of aerodynamic efficiency that varies with α . Lift is required to support weight, drag is produced in the process, and thrust is required to oppose drag. It is normally desirable to maximize this ratio, thus minimizing the thrust and fuel used to offset drag (Section 2.6). The maximum L/D ranges from 2 or less for reentering space vehicles to over 50 for competition sailplanes.

Lift and drag coefficients are typically tabulated as functions of several motion and control variables, such as

$$C_L = C_L(\alpha, M, \delta F) + C_{L_{\hat{q}}}(\alpha, M, \delta F)\hat{q} + C_{L_{\delta E}}(\alpha, M, \delta F)\delta E \quad (2.3-17)$$

$$C_D = C_D(\alpha, M, \delta F) + C_{D_{\hat{q}}}(\alpha, M, \delta F)\hat{q} + C_{D_{\delta E}}(\alpha, M, \delta F)\delta E \quad (2.3-18)$$

where $\hat{q}$ is the normalized pitch rate (Section 3.4),

$$\hat{q} = q\bar{c}/2V \quad (2.3-19)$$

δF is the wing flap setting, and δE is the elevator angle.

The expressions for the aerodynamic rolling, pitching, and yawing moments are similar to the force equations, with the addition of a *reference length* to the dimensionalizing quantities:

$$L = C_l \bar{q} S b \quad (2.3-20)$$

$$M = C_m \bar{q} S \bar{c} \quad (2.3-21)$$

$$N = C_n \bar{q} S b \quad (2.3-22)$$

The reference length is the *wingspan* b for rolling and yawing moments and the *mean aerodynamic chord* $\bar{c}$ for pitching moments. For a wing whose chord length $c(y)$ varies with the spanwise location, y , the *mean aerodynamic chord* (or m.a.c.) of the wing is defined as

$$\bar{c} = \frac{2}{S} \int_0^{b/2} c^2(y) dy \quad (2.3-23)$$

Given a linearly tapered wing with the *taper ratio* λ defined as the tip chord c_T divided by the root chord c_R , the m.a.c. is

$$\bar{c} = \frac{2c_R}{3} \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right) \quad (2.3-24)$$

The aerodynamic coefficients are typically modeled as linear functions of the airflow angles, angular rates, and control settings,

$$C_l = C_{l_o} + C_{l_\beta} \beta + C_{l_{\hat{p}}} \hat{p} + C_{l_{\hat{r}}} \hat{r} + C_{l_{\delta A}} \delta A + C_{l_{\delta R}} \delta R \quad (2.3-25)$$

$$C_m = C_{m_o} + C_{m_\alpha} \alpha + C_{m_{\hat{q}}} \hat{q} + C_{m_{\delta E}} \delta E \quad (2.3-26)$$

$$C_n = C_{n_o} + C_{n_\beta} \beta + C_{n_{\hat{p}}} \hat{p} + C_{n_{\hat{r}}} \hat{r} + C_{n_{\delta A}} \delta A + C_{n_{\delta R}} \delta R \quad (2.3-27)$$

where the coefficients may be functions of flight condition and longitudinal control settings. The normalized roll and yaw rates (Section 3.4) are

$$\hat{p} \triangleq pb/2V \quad (2.3-28)$$

$$\hat{r} \triangleq rb/2V \quad (2.3-29)$$

The aileron, elevator, and rudder deflections for control are represented by δA , δE , and δR .

2.4 Static Aerodynamic Coefficients

The aerodynamic forces and moments of wings, tails, and bodies are central elements of aircraft motion. Here, we are less concerned with the phenomenology than with the gross effects of configuration, motion, and control settings in creating aerodynamic forces and moments. The flow about an entire aircraft is too complex to allow accurate estimates from simple formulas: precise wind tunnel tests or intricate computational fluid dynamics codes are needed for this purpose, and these topics deserve fuller treatment than space allows. Nevertheless, reasonable estimates can be made by adding up the aerodynamic effects of components and accounting for interference effects, using simple formulas to predict the qualitative effects of flight condition. Comprehensive presentations of this handbook approach to estimating aerodynamic coefficients can be found in [F-2, W-7].

Aerodynamic forces are modeled in a general form (exemplified by lift) such as $C_L S \rho V^2/2$, while moments (symbolized by pitch) take the form $C_m S \bar{c} \rho V^2/2$. The reference area and length S and $\bar{c}$ are airplane-specific constants, and it is clear that both forces and moments are expected to follow trends that are quadratic in airspeed V and linear in air density ρ . The application of data from subscale wind tunnel models to estimate full-scale characteristics is predicated on the use of nondimensional coefficients: the full-scale estimates are derived by replacing the model's reference area and length with the full-scale values, adjusting for differences in Reynolds number and other scaling parameters [W-9]. (The *Reynolds number* $Re = V l / \nu$, where ν is the kinematic viscosity [Section 2.1], portrays the relative importance of inertial forces and viscous shear stress in the fluid flow, which govern the thickness and character of laminar [smoothly flowing] or turbulent *boundary layers* about the aircraft.) Consequently, *configuration aerodynamics* focuses on trends in the nondimensional coefficients (e.g., C_L and C_m) rather than the dimensional forces and moments.

Nondimensional aerodynamic coefficients are functions of

- Aircraft shape and proportion
- Aeroelasticity
- Mach and Reynolds numbers
- Airflow angles
- Angular rates
- Control settings

As noted in Section 1.2, the *aircraft configuration* varies greatly with mission and flight regime. Elements that have major impacts on the aerodynamic coefficients include aspect ratio, chord section, leading-edge sweep angle, incidence angle, and twist of the wing, comparable parameters of the horizontal and vertical tail, and fuselage fineness ratio.⁶ Because it normally is generating a force equaling the aircraft's weight, the wing is a major factor in variations of all aerodynamic forces and moments. Air loads proportional to the *dynamic pressure* $\bar{q}$, which is defined as $\rho V^2/2$, bend and twist the airframe; these *aeroelastic distortions* alter the nondimensional coefficients.

The *Mach number* $\mathcal{M} = V/a$ gauges the importance of compressibility effects on airflow, including the formation of shock waves. A body moving through the air creates pressure disturbances that propagate at the speed of sound. As the speed increases, the body catches up with the disturbances propagated forward; hence, the air's compressibility has an important effect on the overall flow. *Shock waves*, which are discontinuities in the pressure field, form when the local flow about the body is accelerated to sonic velocity a or above, that is, when $\mathcal{M} \geq 1$. *Angles of attack* and *sideslip* affect the pressure differentials between windward and leeward surfaces and, therefore, the net forces and moments as well. *Angular rates* produce dynamic effects that are discussed in the next chapter. *Control deflections* reorient aerodynamic surfaces with respect

6. The *aspect ratio* AR is b^2/S , where b is the wingspan and S is the reference area. The *chord section* (or *airfoil*) is the cross-sectional shape of the wing. Principal descriptors are the ratio of maximum thickness to the distance between leading and trailing edges, relative location of the point of maximum thickness, *camber* (curvature) of the mid-thickness line, sharpness of leading and trailing edges, and general shape of the upper and lower wing surfaces. The wing's *sweep angle* Λ is the angle between the y_B axis and the leading edge of the wing. The wing's *incidence angle* i is the angle between the aircraft centerline and a line drawn from the leading to trailing edges of the wing at the root chord. Draw a straight line from the leading to the trailing edge. *Twist* is the net change in the incidence angle of this line (relative to the vehicle centerline) between the root chord and the tip chord. Negative twist (decreasing angle from root to tip) is referred to as *washout*. Assuming a circular cross section, the fuselage fineness ratio relates the length to the maximum diameter.

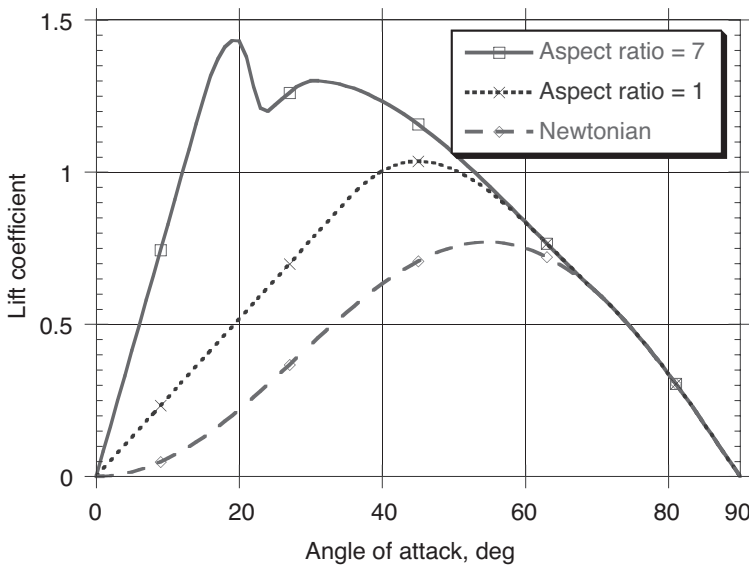


Fig. 2.4-1. Representative wing lift coefficients versus angle of attack.

to the flow, with results that are analogous to changes in the overall air-flow angles. In the remainder of the section, some of the more significant effects of these factors are summarized, with particular attention given to angle of attack, sideslip angle, and Mach number variations.

LIFT

Aerodynamic lift, the force perpendicular to the velocity vector in the aircraft's x_B - z_B plane, is generated primarily by the wing, with contributions from the fuselage, horizontal tail or canard, and engine nacelles. Typically, the wing lift coefficient varies with angle of attack α , as shown for three prototypical cases in Fig. 2.4-1. These curves are loosely representative of business jet and hypersonic transport wing planforms ($AR = 7$ and 1) plus the Newtonian approximation described below. At low α and Mach number, C_L is linear in α [A-3]. It may have a nonzero value (C_{L_0}) at $\alpha = 0$ deg due to the wing and tail incidence relative to the centerline, airfoil camber (curvature of the mean chord line between upper and lower surfaces), spanwise twist of the chord section, and fuselage shape. The lift coefficient necessarily reaches a maximum value $C_{L_{max}}$ at some α between 0 and 90 deg for reasons of flow geometry and separation.

Consider a simple geometrical example based on the *Newtonian flow* model [A-3]. An infinitesimally thin flat plate of area S is immersed in an

airflow with density and speed of ρ and V . The *edge-on case* is represented by $\alpha = 0$ deg and the *broadside condition* by $\alpha = 90$ deg. Newton concluded (from his second law) that a flow uniformly brought to rest by a perpendicular surface ($\alpha = 90$ deg) exerts a normal force N on the surface of

$$\begin{aligned} N &= (\text{mass flow rate per unit area})(\text{change in velocity})(\text{area of surface}) \\ &= (\rho V)(V)(S) = \rho V^2 S \end{aligned} \quad (2.4-1)$$

Modeling this force as the product of a nondimensional coefficient, dynamic pressure, and reference area,

$$N \triangleq C_N \bar{q} S = C_N \left(\frac{\rho V^2}{2} \right) S \quad (2.4-2)$$

leads to the conclusion that $C_N = 2$. This judgment overlooks air viscosity and important details of the flow, and it ignores the leeward surface entirely, yet it produces a useful working hypothesis for some flow regimes. For angles less than 90 deg, the mass flow rate intercepted by the plate would be reduced to $\rho V \sin \alpha$, while the change in flow velocity would be $V \sin \alpha$; hence, the nondimensional coefficient expressing the normal force per unit area on the plate (i.e., the differential pressure) would be

$$C_N = C_{N_{max}} \sin^2 \alpha = 2 \sin^2 \alpha \quad (2.4-3)$$

The plate is at angle α to the flow, so the corresponding lift coefficient is

$$C_L = C_N \cos \alpha = 2 \sin^2 \alpha \cos \alpha \quad (2.4-4a)$$

as plotted in Figure 2.4-1 (Newtonian). This function equals zero and has zero slope at zero angle of attack, and the lift coefficient's maximum value is 0.77 at $\alpha = 54.74$ deg. At higher α , C_N is larger, but the force component perpendicular to the flow (i.e., C_L) is smaller.

Equation 2.4-4a represents trends in the lift coefficient of all aircraft configurations with *plan area*⁷ S_p in two circumstances:

- Large angle of attack (>60 deg) at all Mach numbers
- All angles of attack at hypersonic Mach numbers ($M > 5$)

The fit to real data can often be improved by choosing $C_{N_{max}}$ other than 2, as well as by accounting for any differences between the reference and plan areas. Thus, eq. 2.4-4a would be modified to be

$$C_L = C'_{N_{max}} \left(\frac{S_p}{S} \right) \sin^2 \alpha \cos \alpha \quad (2.4-4b)$$

7. In this usage, the *plan area* is the horizontal area of the entire aircraft projected onto the x_B, y_B plane, including wing, fuselage, and horizontal tail.